18.09.2025

Ontologies, Knowledge Graphs, and Neuro-symbolic AI

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Outline

1. Ontologies and KG in the semantic web
2. Semantic Frames as possible conceptual modeling approach
3. Applied Ontology Use Case

photo by Evan Vucci/The Associated Press

Semantic Web

World Wide Web extension composed of:

- semantic relations, machine-understandable

The overall idea is:

- a universe of machine-understandable data:
    - for intelligent agents to interpret and understand data on the Web
    - to perform machine inference

To do this: semantics!

RDF - Resource Description Framework

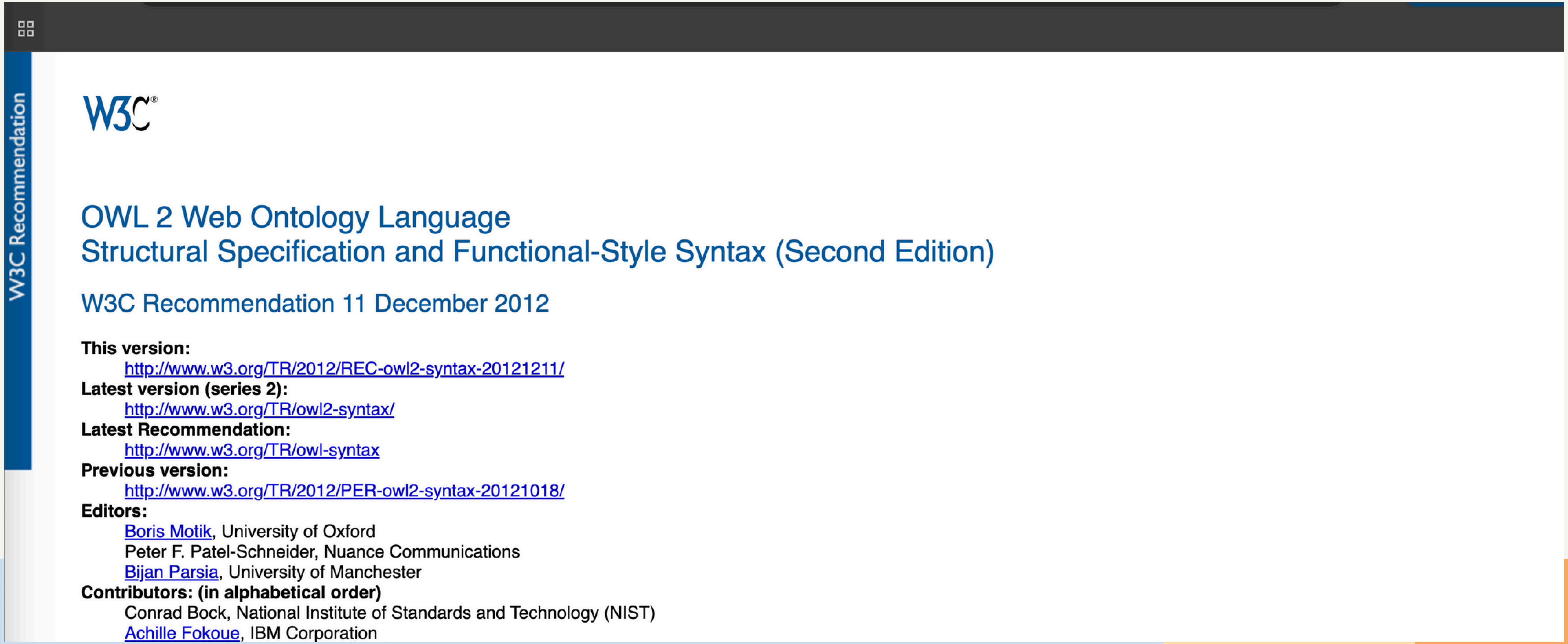
Fundamental principles:

1. Anything can be identified by an IRI
2. Something can be said about anything
3. Simple language to define anything

IRI: Internationalized Resource Identifier
It is built with the following structure:

protocol://domain/path
https://w3id.org/framester/framenet/abox/frame/Event

OWL Language



W3C Recommendation

W3C®

OWL 2 Web Ontology Language
Structural Specification and Functional-Style Syntax (Second Edition)

W3C Recommendation 11 December 2012

This version:
http://www.w3.org/TR/2012/REC-owl2-syntax-20121211/
Latest version (series 2):
http://www.w3.org/TR/owl2-syntax/
Latest Recommendation:
http://www.w3.org/TR/owl-syntax
Previous version:
http://www.w3.org/TR/2012/PER-owl2-syntax-20121018/
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Contributors: (in alphabetical order)
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Achille Fokoue, IBM Corporation

https://www.w3.org/TR/owl2-syntax/

Turtle syntax

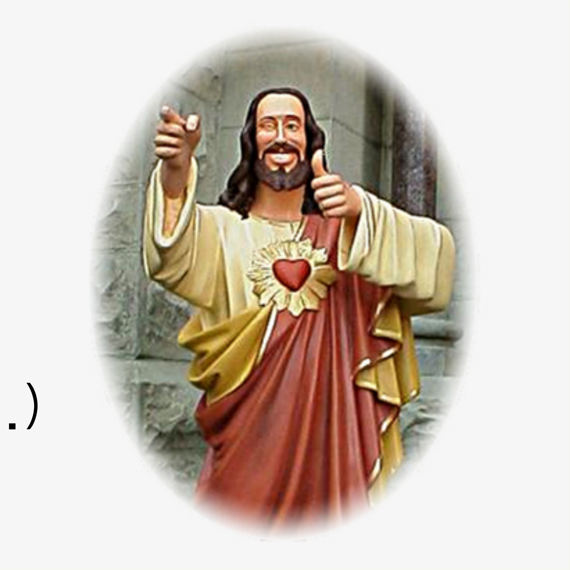
```
@base <http://example.org/> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix foaf: <http://xmlns.com/foaf/0.1/> .
@prefix dolcebasic: <https://w3id.org/DOLCE/OWL/DOLCEbasic/>

:sdg :tutorAt :isao2025 ;
	a dolcebasic:AgentivePhisicalObject ;
	foaf:name "Stefano De Giorgis"
```

https://www.w3.org/TR/turtle/

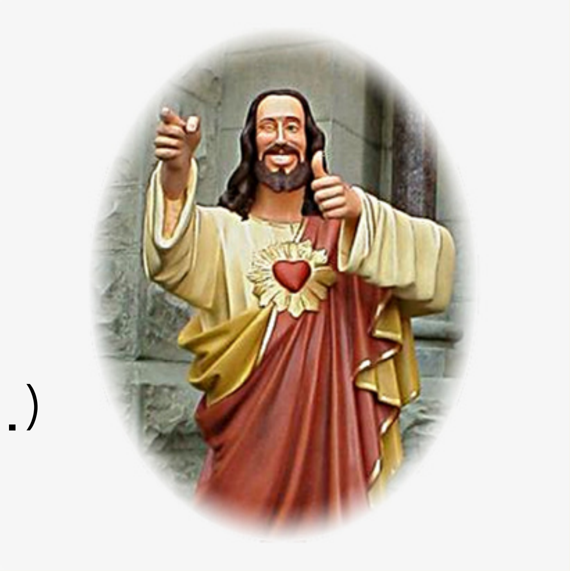
Why ontologies?

Why ontologies?



- Investigating the nature of a domain / entity / phenomenon (sometimes we just play being God.)

Why ontologies?



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- Shared conceptualisations allow heterogeneous systems to exchange and interpret data consistently (across the Web)

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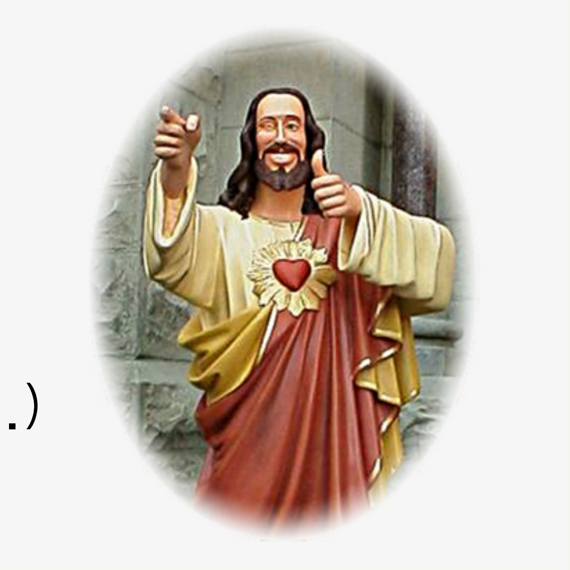

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- We need to align different data models or conceptual schemas

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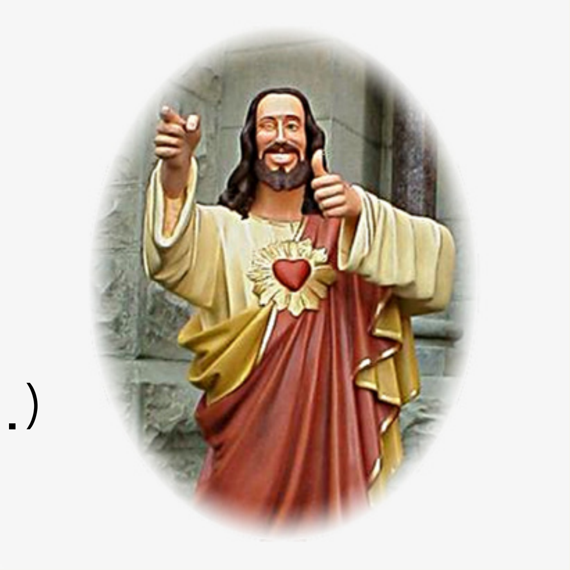

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- We have heterogeneous types of data and we want to include them all in the same knowledge system

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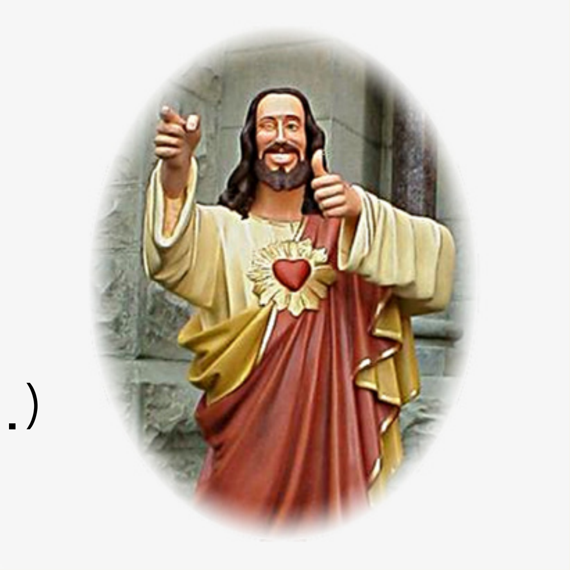

- Investigating the nature of a domain / entity / phenomenon (sometimes we just play being God.)
- Shared conceptualisations allow heterogeneous systems to exchange and interpret data consistently (across the Web)
- We need to align different data models or conceptual schemas
- We have heterogeneous types of data and we want to include them all in the same knowledge system
- For machines to do *predictable inference*

OPEN KNOWLEDGE GRAPHS / ONTOLOGIES

- DBpedia
- WikiData
- BabelNet
- FreeBase
- YAGO
- ConceptNet
- Framester
- Gene Ontology
- ...

ENTERPRISE KNOWLEDGE GRAPHS / ONTOLOGIES

KNOWLEDGE GRAPHS EXAMPLES

One possible conceptual approach: Frame Semantics

Semantic Frame

Originally from Fillmore (Fillmore et al., 2006) in the cognitive linguistics field, extends the Case Grammar.

The intuition was previously already used in AI by Minsky (Minsky et al., 1974) and Brachman (Brachman, 1977).

Intuitively, a semantic frame is a schematic representation of a the abstract structure of a situation, based on identifying a set of roles.

I love you.

I love you. $\rightarrow$ x loves y

$\exists x \exists y \mathsf{L}(x, y)$

I love you. → x loves y

$\exists x \exists y \mathsf{L}(x, y)$

→ x cares about y
(cognitive/moral)

I love you. → x loves y

$\exists x \exists y L(x, y)$

x would behave in order not to harm y (behavioral)

x cares about y (cognitive/moral)

x has a range of neurophysiological manifestations when in the presence of/thinking about y

I love you.

x loves y

$\exists x \exists y L(x, y)$

x would behave in order not to harm y (behavioral)

x cares about y (cognitive/moral)

How many arguments?

```
loves(x)
loves(x,y)
loves(x,y,z,...n)
```

x has a range of neurophysiological manifestations when in the presence of/thinking about y

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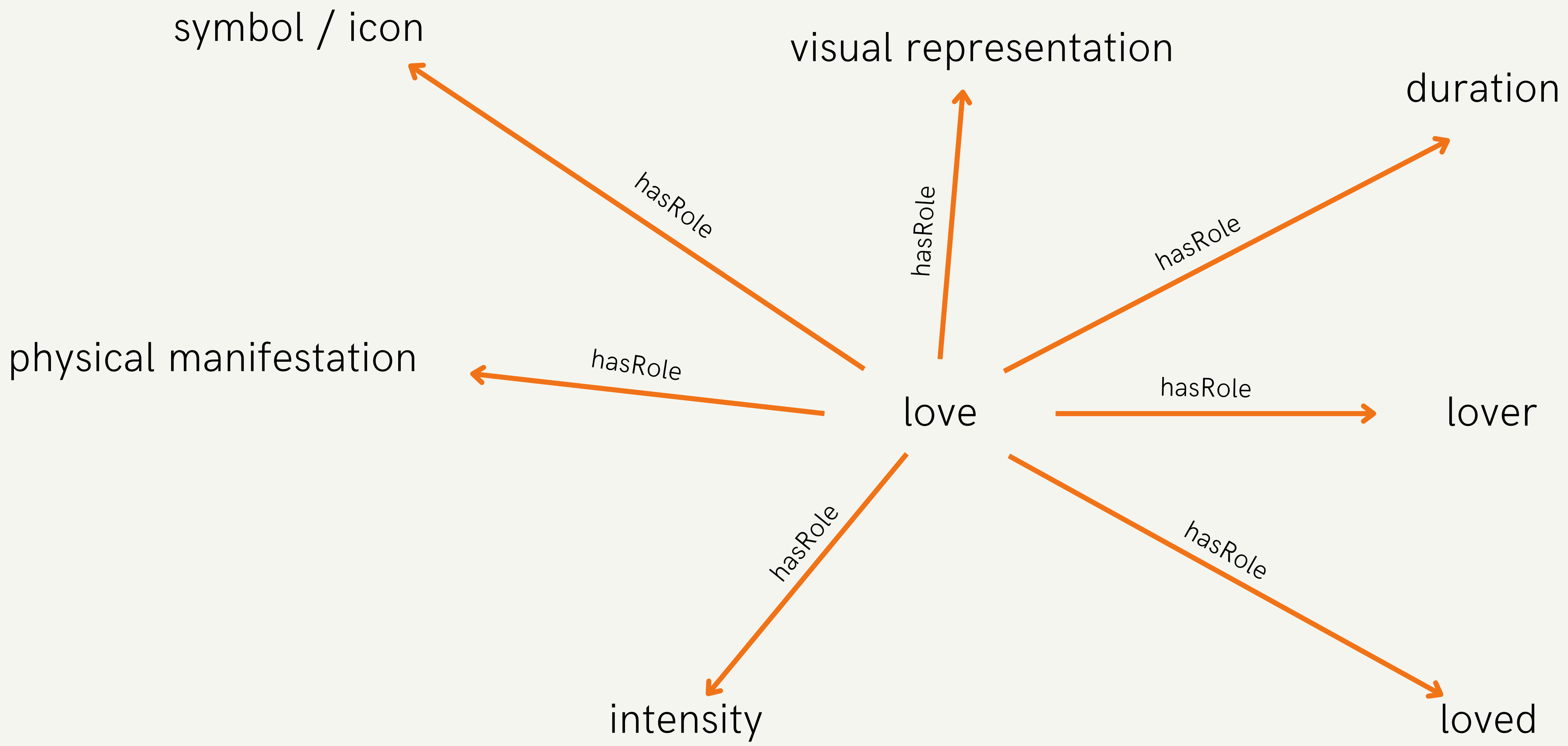
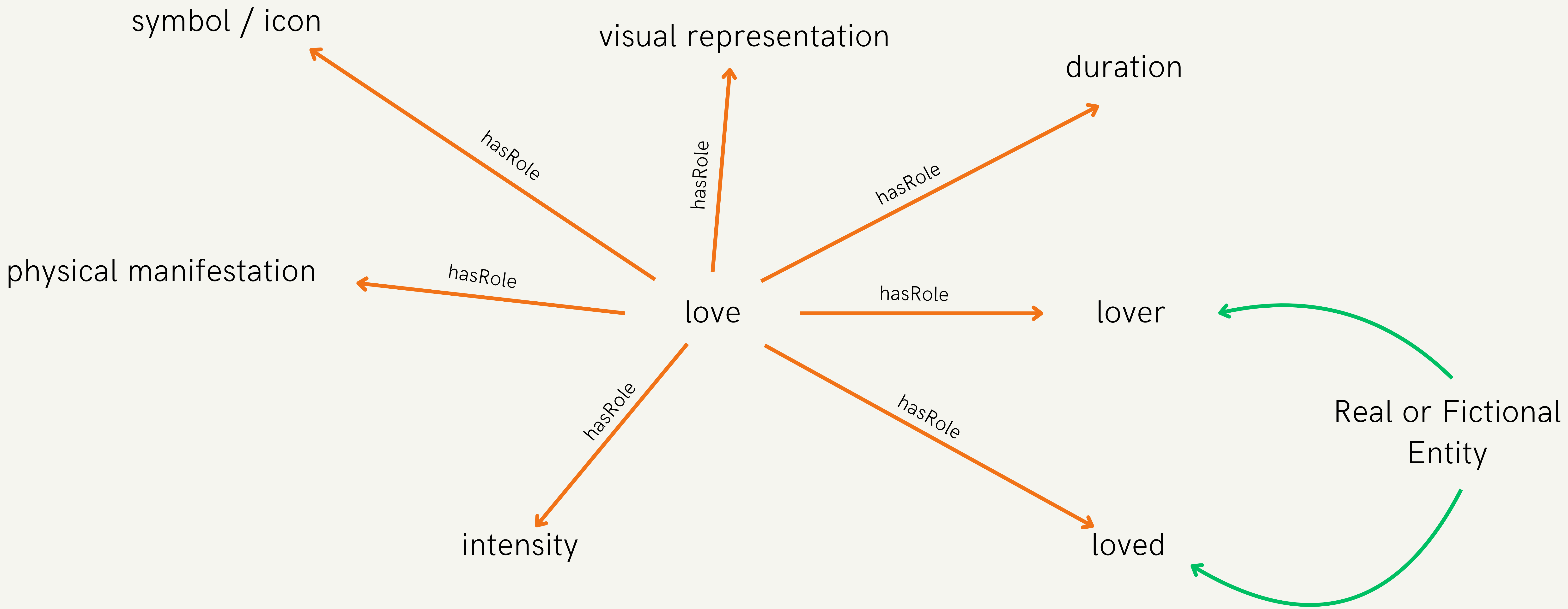
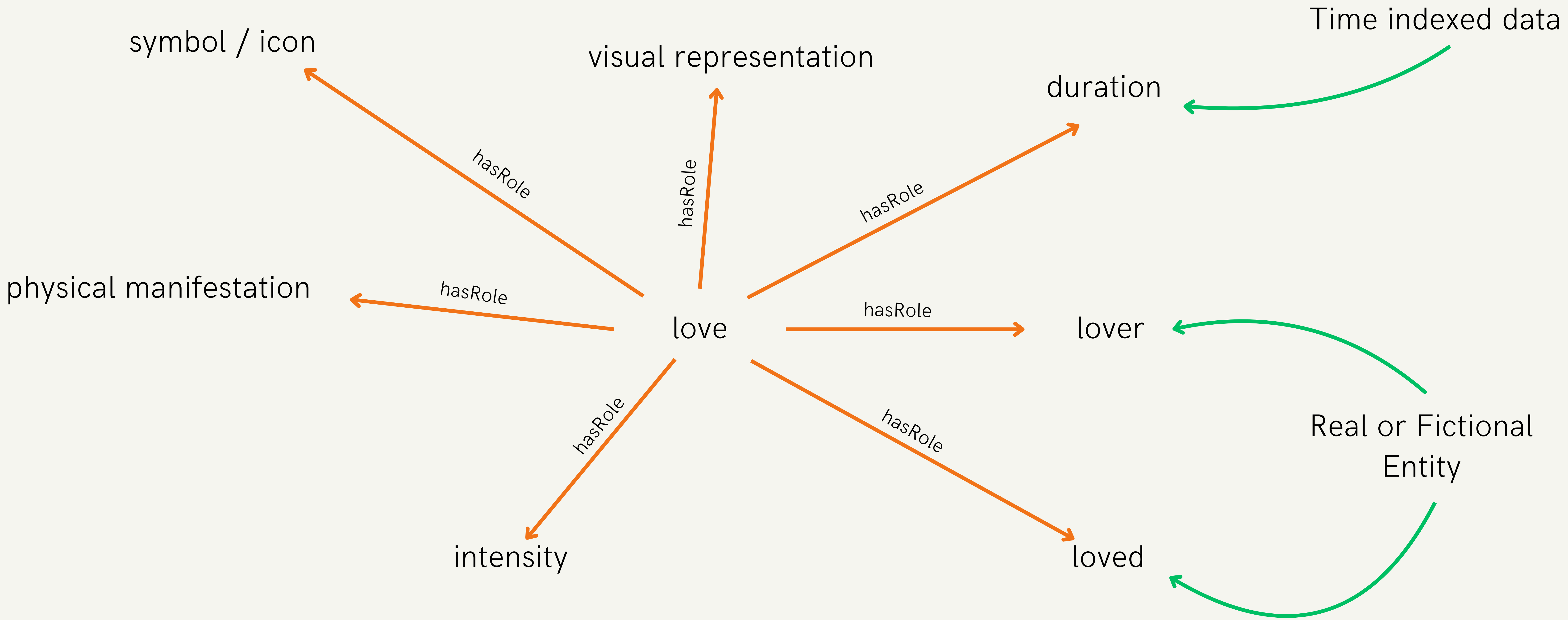


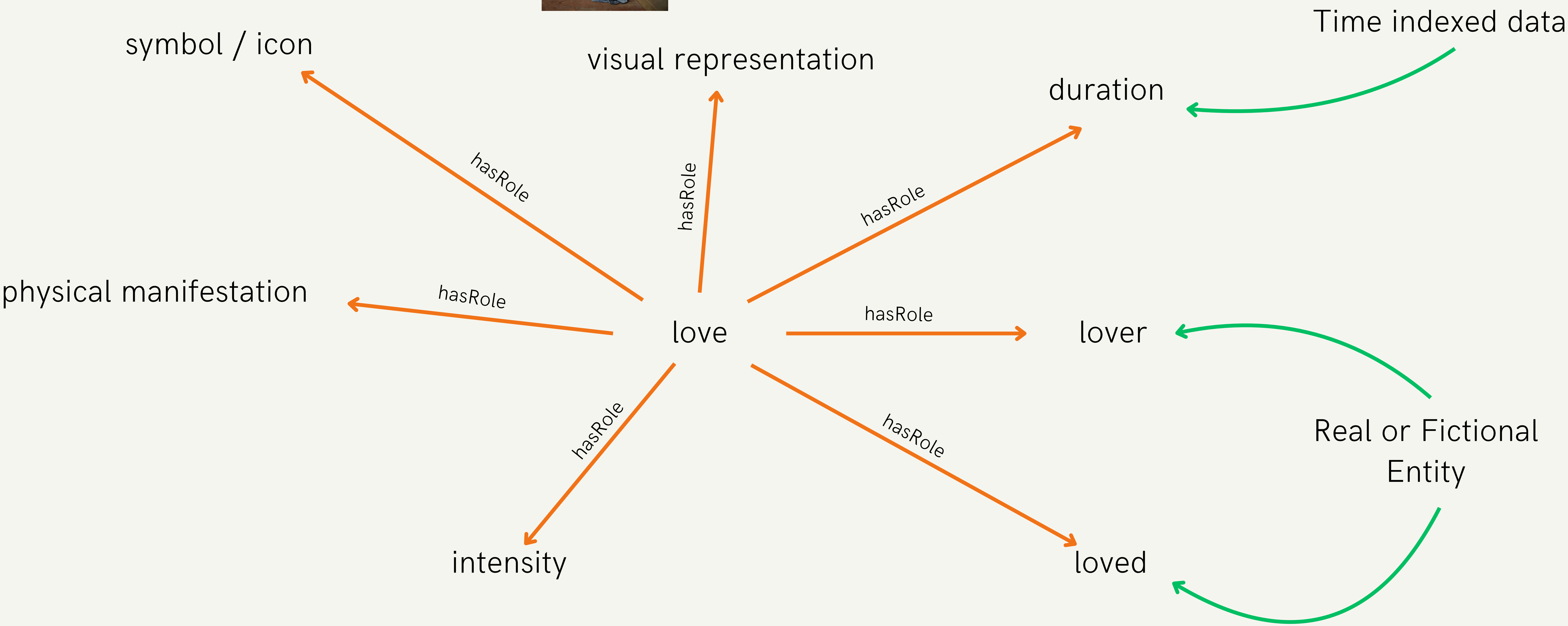
"LOVE" SEMANTIC FRAME



"LOVE" SEMANTIC FRAME



"LOVE" SEMANTIC FRAME



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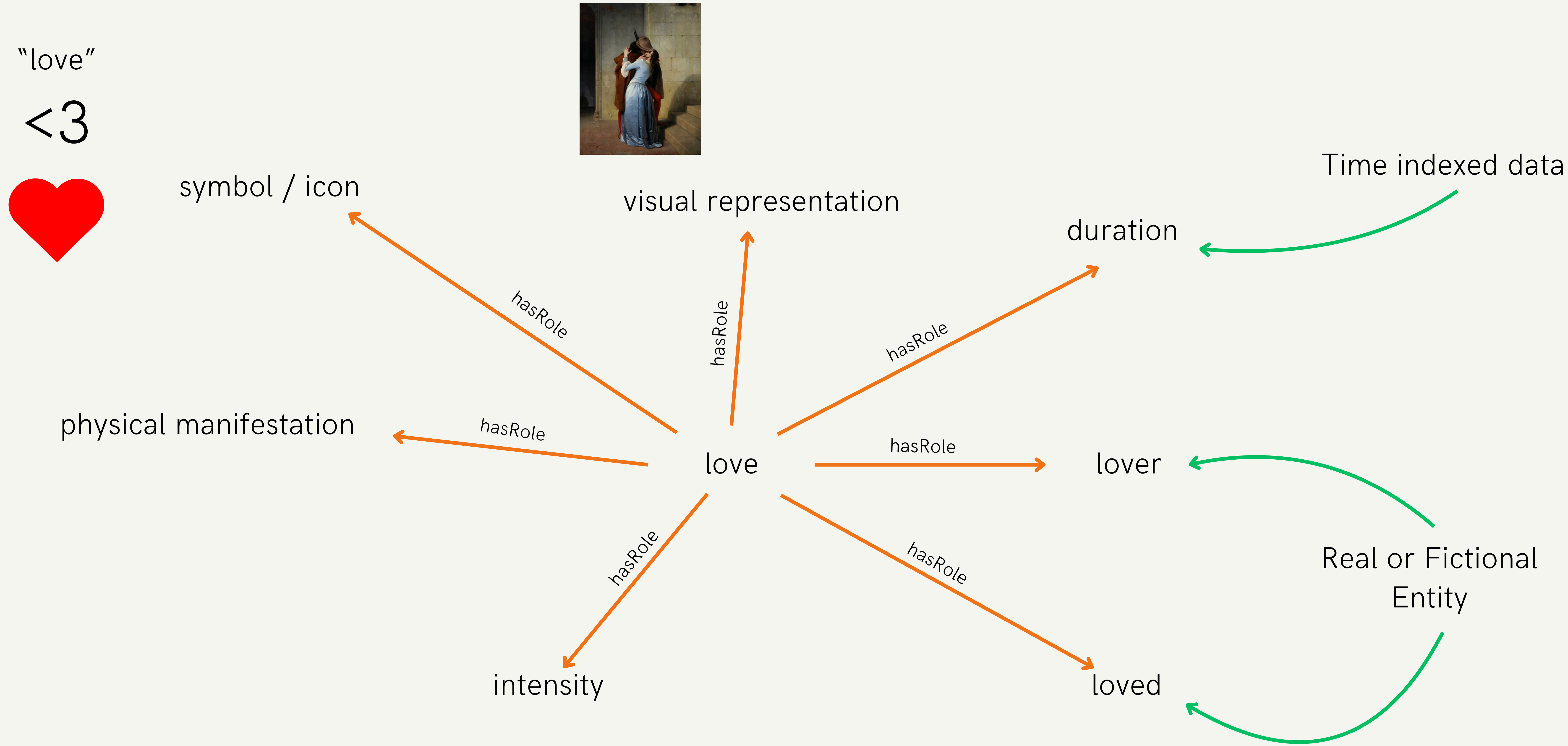
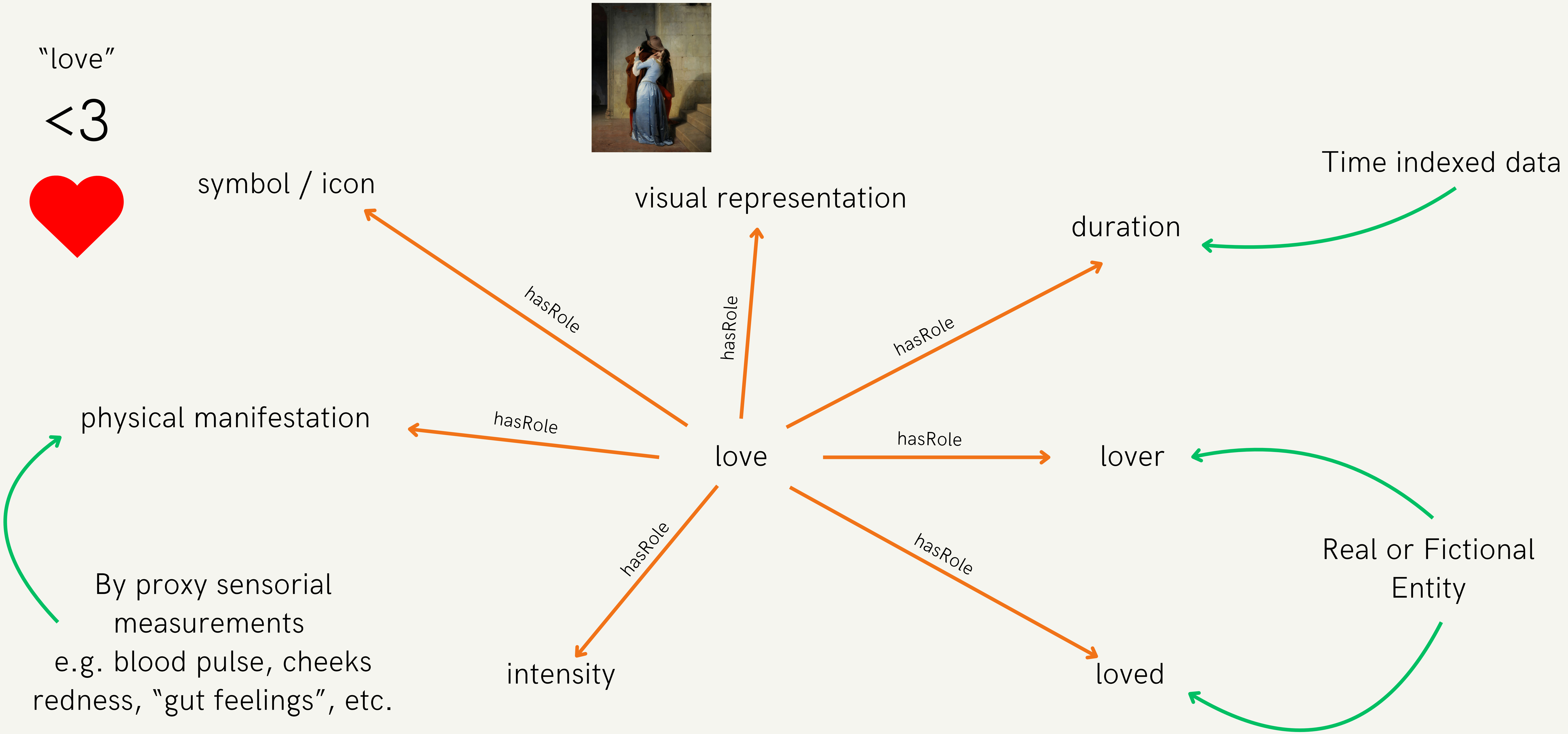
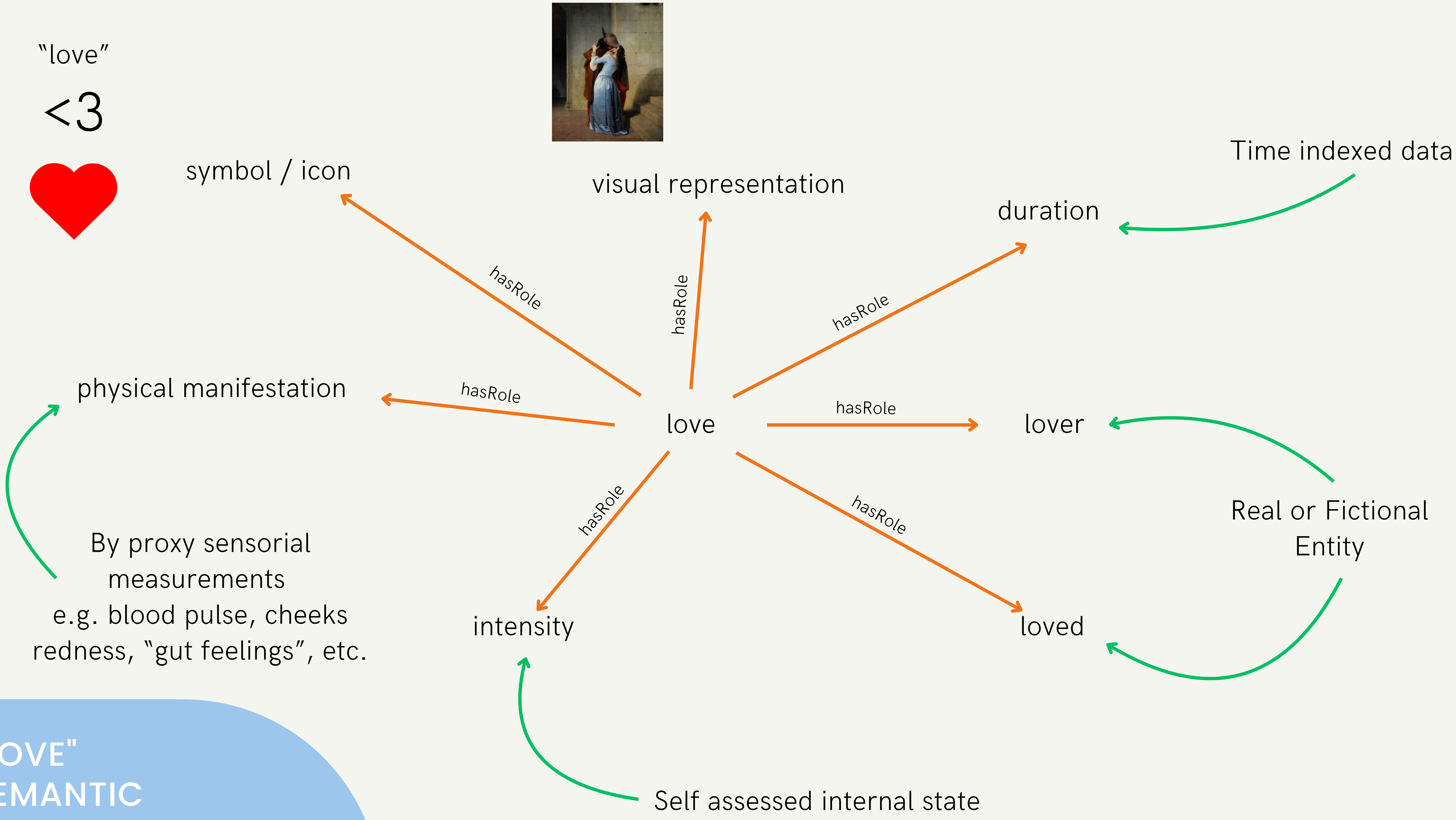


"LOVE" SEMANTIC FRAME



"LOVE" SEMANTIC FRAME



"LOVE" SEMANTIC FRAME

Frame structure in RDF

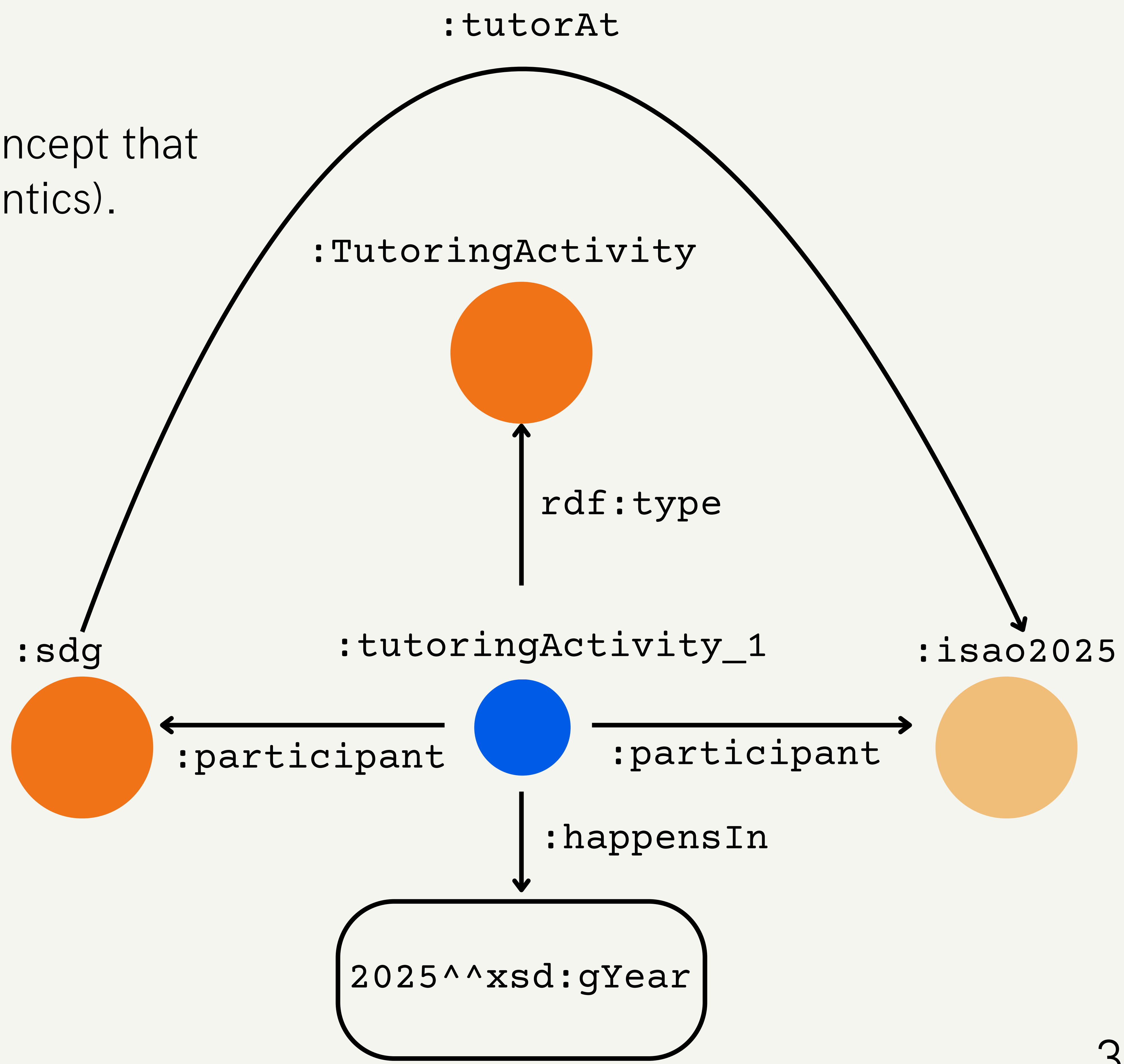
N-ary relations

It models a situation via a new relationship concept that connects all arguments (close to Frame Semantics).

Stefano is a tutor at ISAO 2025.

```
:sdg :tutorAt :isao2025 .
```

```
:tutoringActivity_1 rdf:type
:TutoringActivity ;
   :participant :sdg ;
   :participant :isao2025 ;
   :happensIn "2025"^^xsd:gYear .
```



Frame-based resources

- FrameNet
- PropBank
- VerbNet

Attack

Lexical Unit Index

Definition:

An Assailant physically attacks a Victim (which is usually but not always sentient), causing or intending to cause the Victim physical damage. A Weapon used by the Assailant may also be mentioned, in addition to the usual Place, Time, Purpose, Explanation, etc. Sometimes a location is used metonymically to stand for the Assailant or the Victim, and in such cases the Place FE will be annotated on a second FE layer.

As soon as he stepped out of the bar he was SET upon by four men in ski-masks.

Is he INVADING Iraq just to cover other shortcomings?

Then Jon-O's forces AMBUSHED them on the left flank from a line of low hills.

FEs:

Core:

Assailant [Asl] **Semantic Type:** Sentient	The person (or other self-directed entity) that is attempting physical harm to the Victim. The mysterious fighter ATTACKED the guardsmen with a sabre.
Victim [Vic] **Semantic Type:** Sentient	This FE is the being or entity that is injured by the Assailant's attack. The mysterious fighter ATTACKED the guardsmen with a sabre.

```
<frameset>
  <predicate lemma="attack">
    <roleset id="attack.01" name="to make an attack, criticize strongly">
      <aliases>
        <alias pos="n">attacking</alias>
        <alias pos="n">attack</alias>
        <alias pos="v">attack</alias>
        <argalias arg="0" pos="n">attacker</argalias>
      </aliases>
      <roles>
        <role descr="attacker" f="PAG" n="0">
          <rolelinks>
            <rolelink class="judgment-33.1-1" resource="VerbNet" version="verbnet3.3">agent</rolelink>
            <rolelink class="Judgment_communication" resource="FrameNet" version="1.7">communicator</rolelink>
            <rolelink class="Attack" resource="FrameNet" version="1.7">assailant</rolelink>
            <rolelink class="judgment-33.1-1" resource="VerbNet" version="verbnet3.4">agent</rolelink>
            <rolelink class="attack-60.1-1" resource="VerbNet" version="verbnet3.4">agent</rolelink>
          </rolelinks>
```

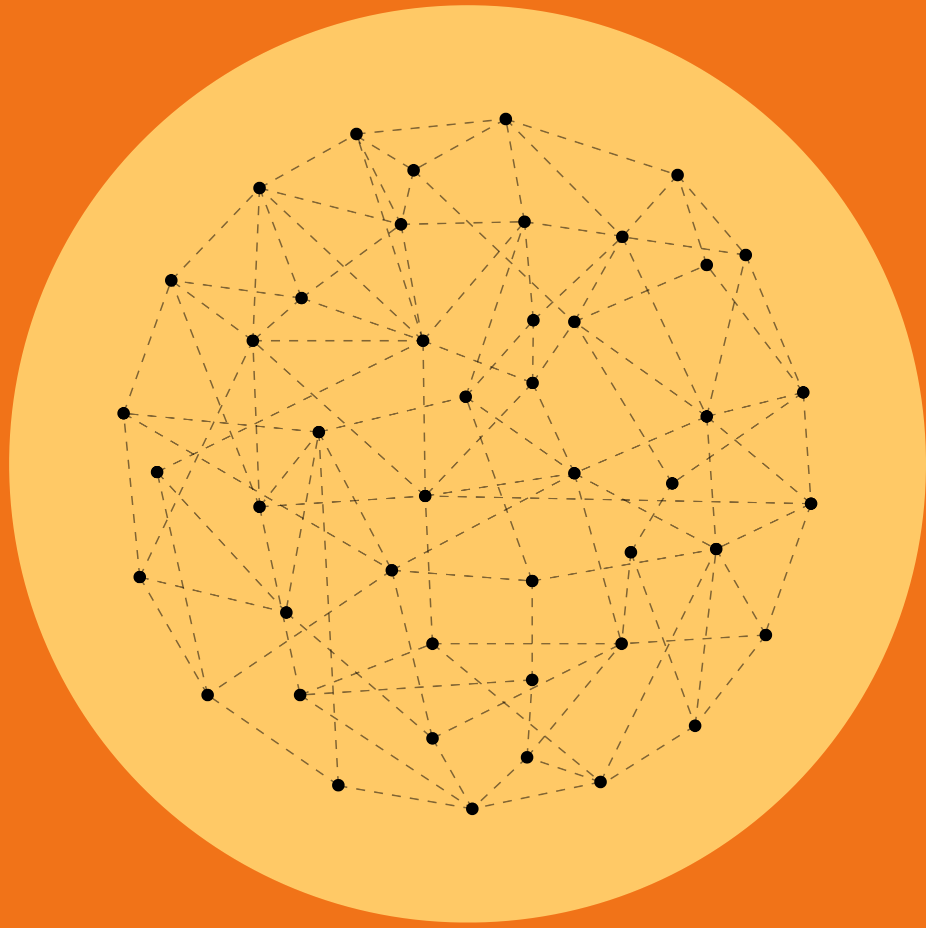
master verbnet / verbnet3.4 / attack-60.1.xml

Blame

```
<THEMROLES>
 <THEMROLE type="Agent">
  <SELRESTRS/>
 </THEMROLE>
 <THEMROLE type="Patient">
  <SELRESTRS/>
 </THEMROLE>
</THEMROLES>
```

photo by Evan Vucci/The Associated Press

Neuro-symbolic AI



Neuro-symbolic AI

1. Symbolic Neuro symbolic
2. Symbolic[Neuro]
3. Neuro | Symbolic
4. Neuro: Symbolic → Neuro
5. Neuro_{Symbolic}
6. Neuro[Symbolic]

The third AI summer: AAAI Robert S. Engelmore Memorial Lecture

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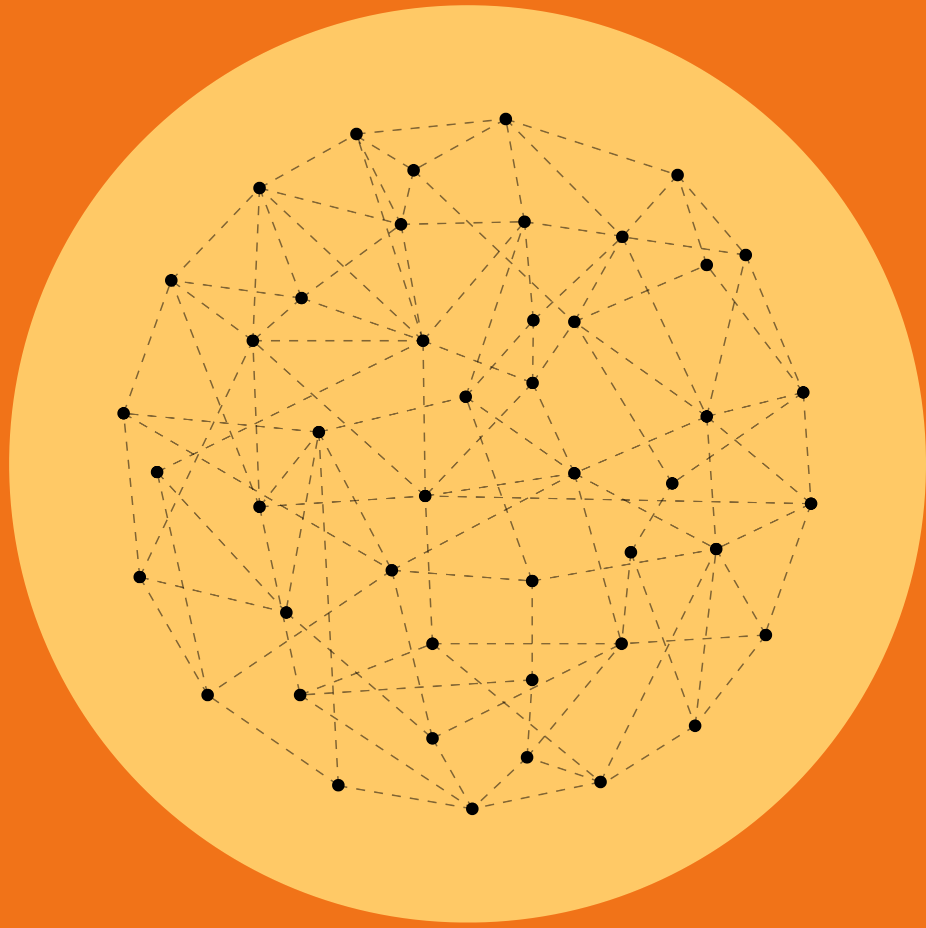
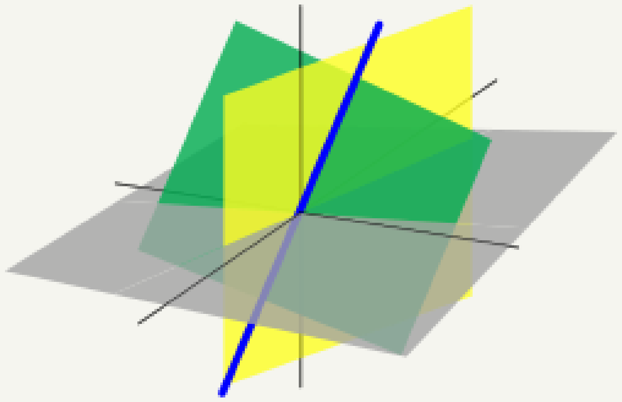
Correspondence
Henry A. Kautz, Department of Computer Science, University of Rochester, Rochester, NY 14627, USA.
Email: henry.kautz@gmail.com



1. Symbolic Neuro symbolic

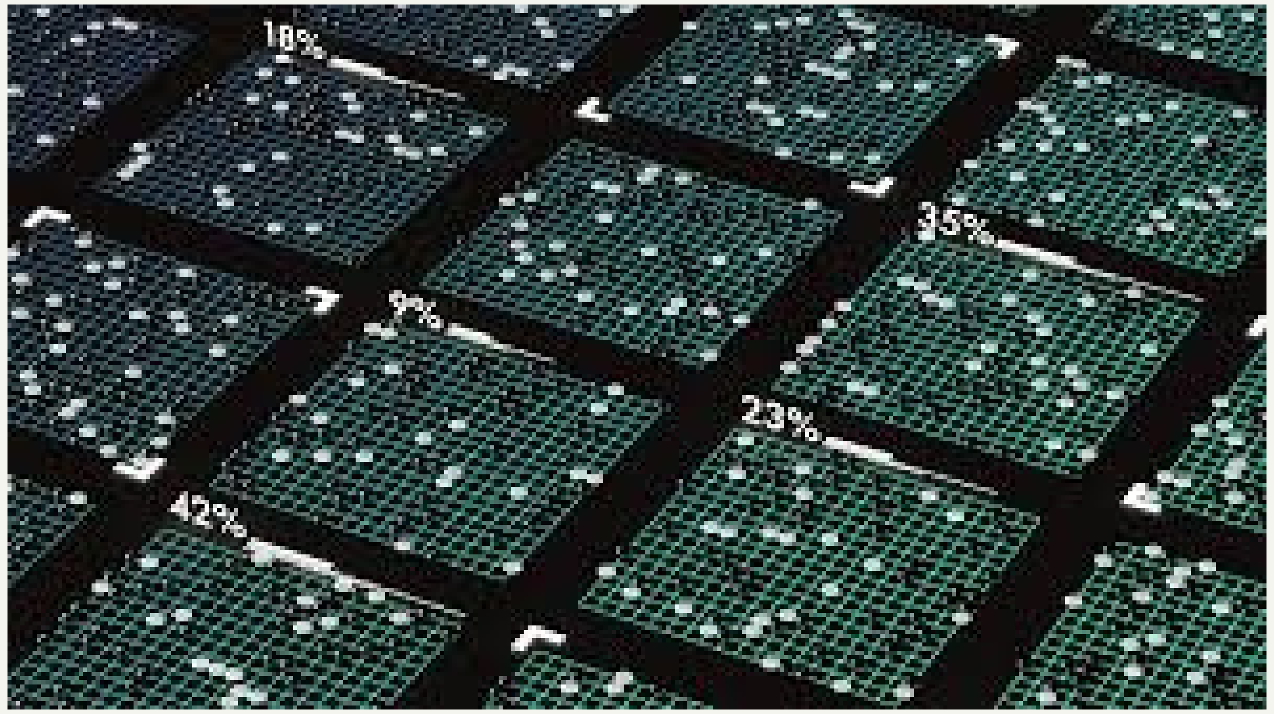
Natural language text

e.g. Word2Vec, GloVe, etc.



Natural language text

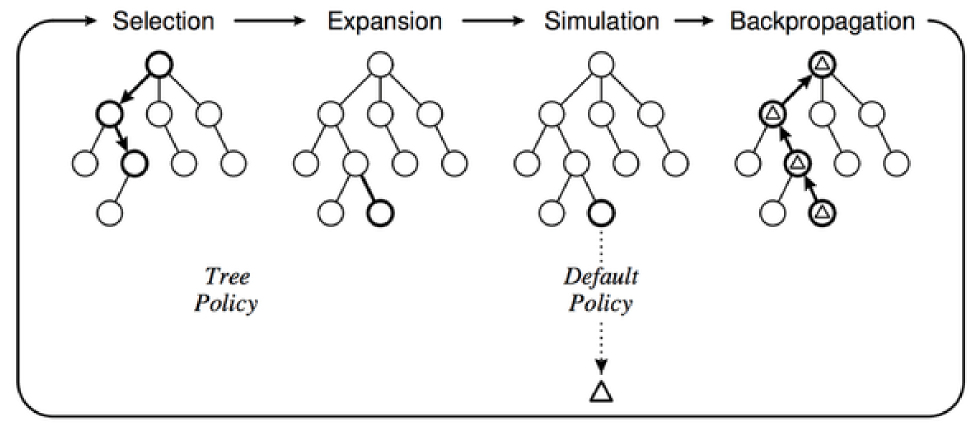
2. Symbolic[Neuro]



Alpha Go

Symbolic Problem Solver

Monte Carlo tree search



+

Evaluation Function

Neural Network

3. Neuro | Symbolic

Visual Question Answering

YOLO v8

"How many people are there in the image?"

Symbolic reasoner counting the entities.

Answer: ...

4. Neuro: Symbolic → Neuro

Mathematical Expression Semplification
[Lample and Charton (2020)]

$$2x + 4(5 + 2) + 3^2 - (3 + 4/2) =$$
$$\mathbf{= 2x + 32}$$

Transformers Seq2Seq
Deep Learning System

5. Neuro_{Symbolic}

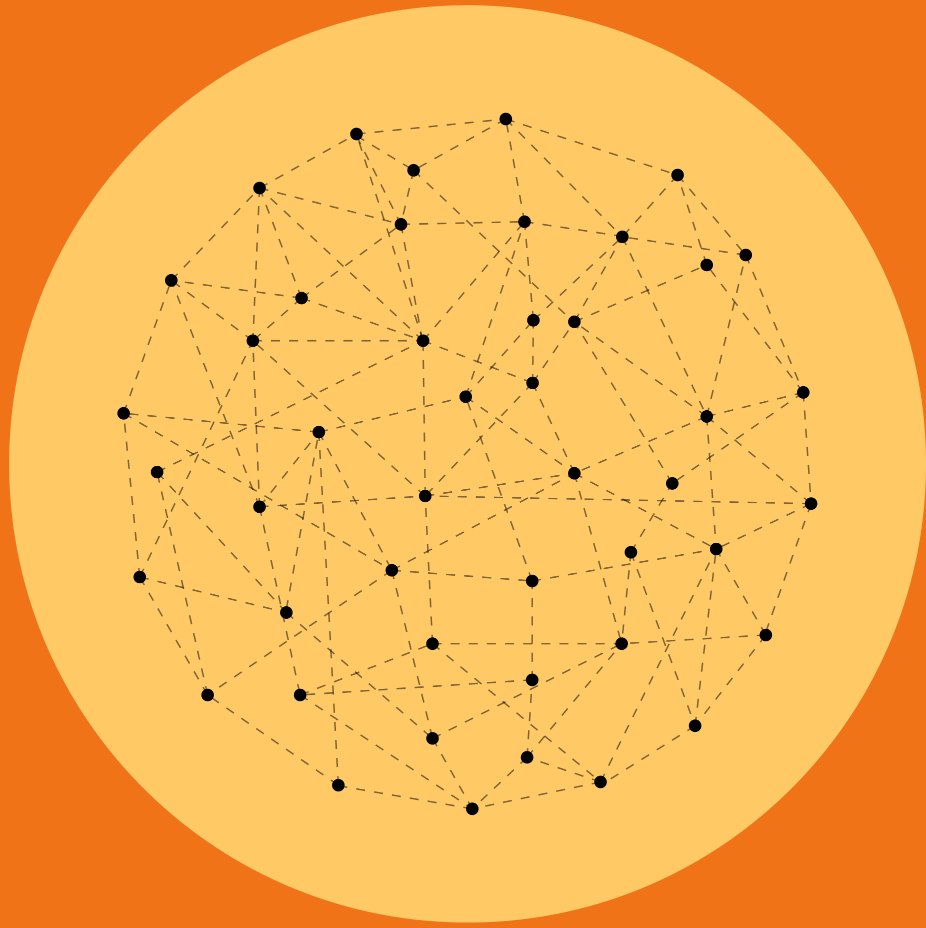
- Knowledge Graph Embeddings: Symbolic relationships (parent-of, capital-of) are embedded as geometric relationships in neural space
- Physics-Informed Neural Networks: Physical laws are built into the network architecture as constraints
- Hierarchical Classification: Tree-structured categories are embedded in the network's layer organization

5. Neuro_{Symbolic}

The idea is to embed a symbolic reasoning engine inside a neural engine.

It resonates with Kahneman "system 1 and system 2", but it is still highly underdeveloped.

Neuro-symbolic knowledge enrichment - 1



FRAMENET


photo by Evan Vucci/The Associated Press

The assassination attempt on Donald Trump.

photo by Evan Vucci/The Associated Press

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photo by Evan Vucci/The Associated Press

The assassination attempt on Donald Trump.

(~Without Background Knowledge)

A political figure or leader, doing a victory gesture, surrounded by guards / agents protecting him. Some blood spilling from his ear, an American flag on the background.

photo by Evan Vucci/The Associated Press

The assassination attempt on Donald Trump.

(~Without Background Knowledge)

A political figure or leader, doing a victory gesture, surrounded by guards / agents protecting him. Some blood spilling from his ear, an American flag on the background.

(Scene Description ~Only)

A man in an elegant suit in the center, with a raised arm, surrounded by some people gathering around his body. Some blood spilling from his ear, an American flag on the background.

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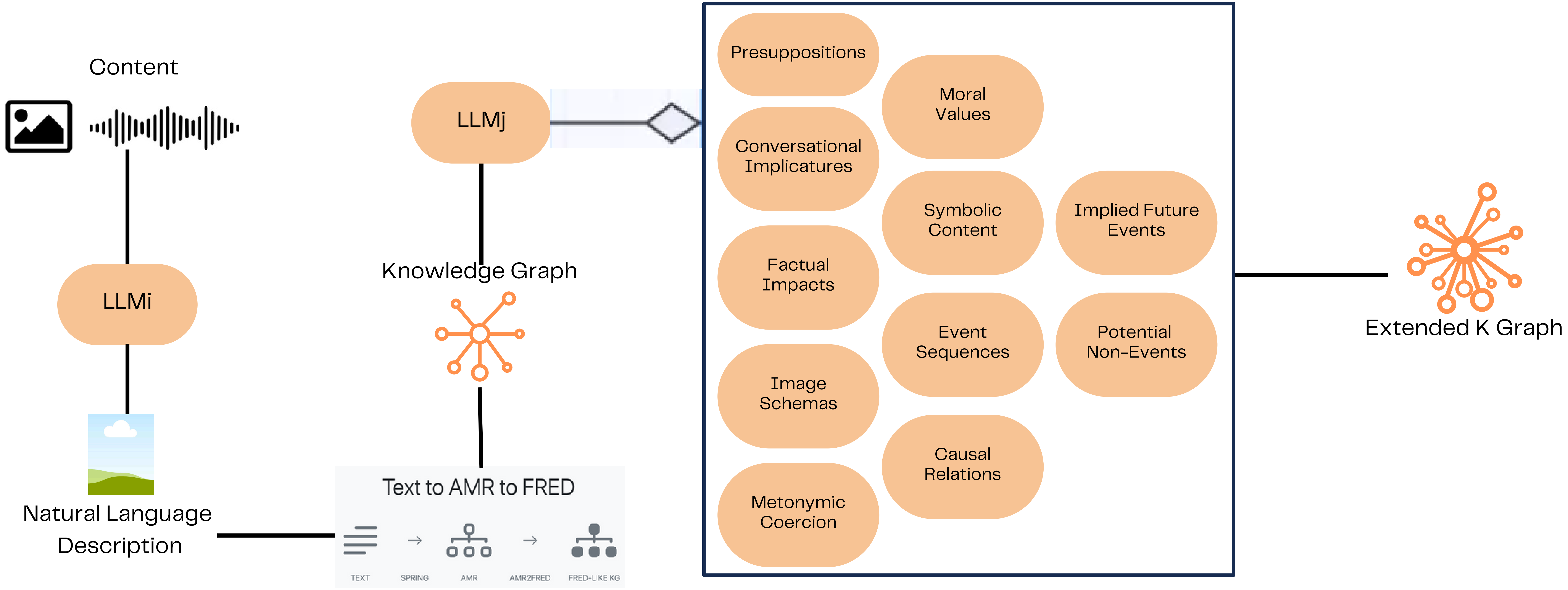
Knowledge Extension Pipeline



Off-the-shelf formal graph extensions from automated LLM prompting (11 implicit motifs)

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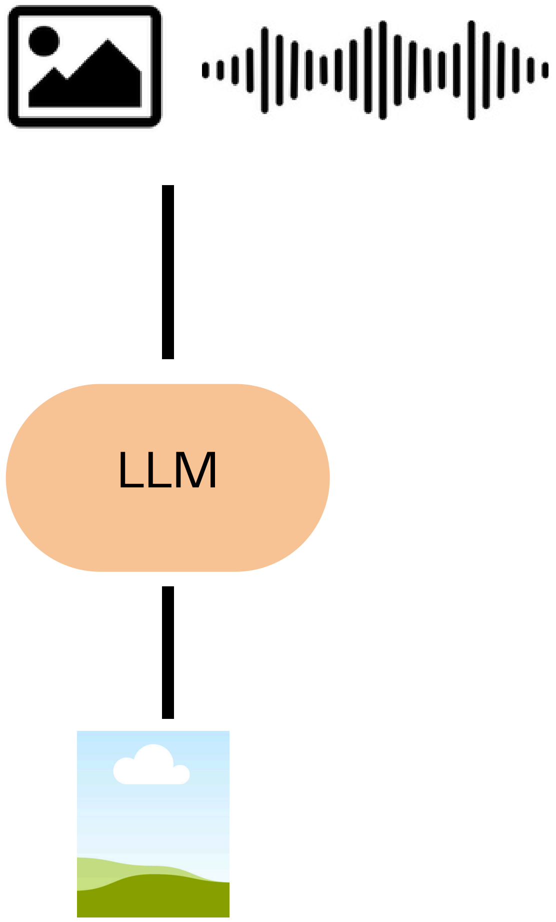
Knowledge Extension Pipeline - 2



Natural Language Description

Possible Clever Prompts:

1. *Describe the overt, implicit and symbolic content of the attached image.*
2. *Describe this photo in detail, including gestures, participants, public roles, situations, bodily position, and general implied meaning.*
3. ...

```
A high-profile political figure raises his fist in
defiance, surrounded by vigilant security personnel during
a public rally, with a large American flag prominently
displayed in the background, emphasizing themes of
patriotism and protection.
```

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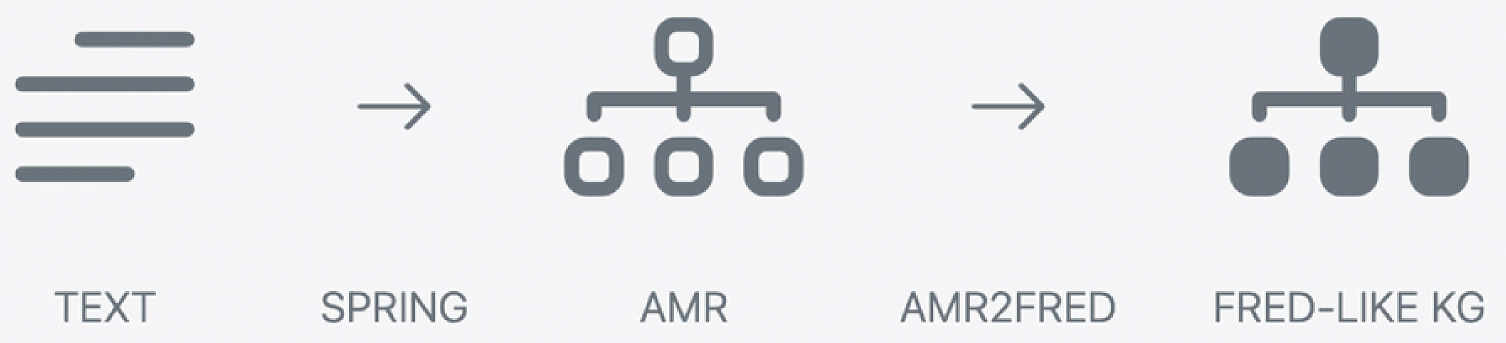
Text to AMR to FRED



Test it!

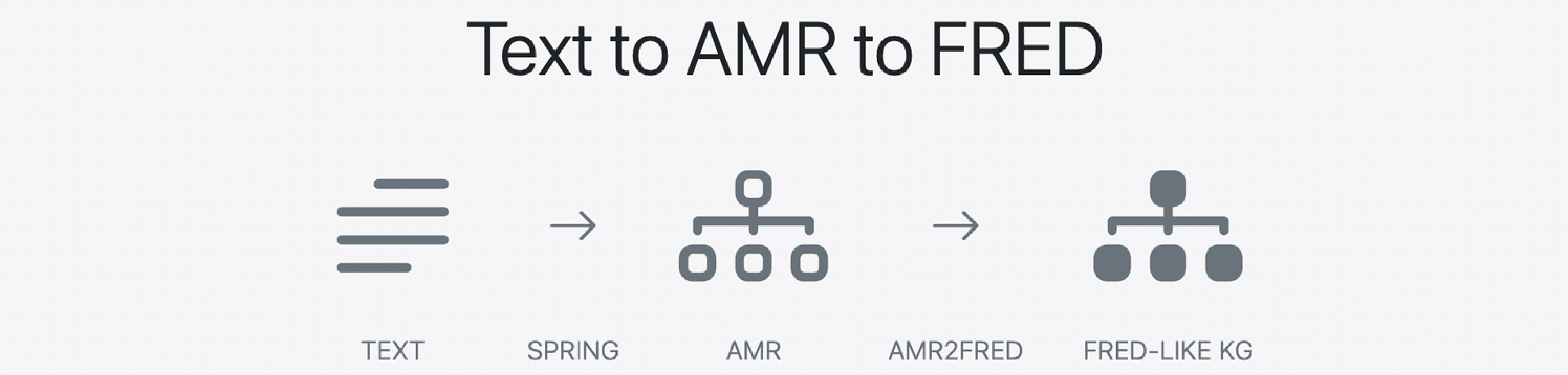
Given some text as input, this tool will parse it into an AMR (Abstract Meaning Representation) graph, using SPRING. The AMR graph is then converted into an RDF/OWL knowledge graph that follows FRED's knowledge representation patterns, using AMR2FRED.

A high-profile political figure raises his fist in defiance, surrounded by vigilant security personnel during a public rally, with a large American flag prominently displayed in the background, emphasizing themes of patriotism and protection.

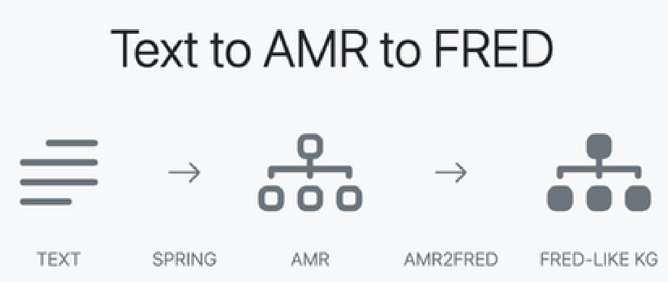
Settings

Convert text

https://arco.istc.cnr.it/itaf/



A high-profile political figure raises his fist in defiance, surrounded by vigilant security personnel during a public rally, with a large American flag prominently displayed in the background, emphasizing themes of patriotism and protection.



AMR excerpt due to its dimension

```
(z0 / raise-01
    :ARG0 (z1 / figure
              :mod (z2 / politics)
              :mod (z3 / profile
                       :ARG1-of (z4 / high-02))
              :ARG1-of (z5 / surround-01
                           :ARG2 (z6 / personnel
                                     :mod (z7 / security)
                                     :ARG0-of (z8 / vigilant-01)))
              :ARG0-of (z9 / have-03
                           :ARG1 (z10 / flag
                                      :mod (z11 / large)
                                      :mod (z12 / country
                                                :wiki "United_States"
                                                :name (z13 / name
                                                           :op1 "America"))
                             ...
```

Text to AMR

Your text

PENMAN | Graph

This is the result of converting your text to AMR using SPRING. The AMR graph is serialized using the PENMAN notation.

```
(z0 / raise-01
    :ARG0 (z1 / figure
              :mod (z2 / politics)
              :mod (z3 / profile
                       :ARG1-of (z4 / high-02))
              :ARG1-of (z5 / surround-01
                           :ARG2 (z6 / personnel
                                     :mod (z7 / security)
                                     :ARG0-of (z8 / vigilant-01)))
              :ARG0-of (z9 / have-03
                           :ARG1 (z10 / flag
                                      :mod (z11 / large)
                                      :mod (z12 / country
                                                :wiki "United_States"
                                                :name (z13 / name
                                                           :op1 "America"))
```

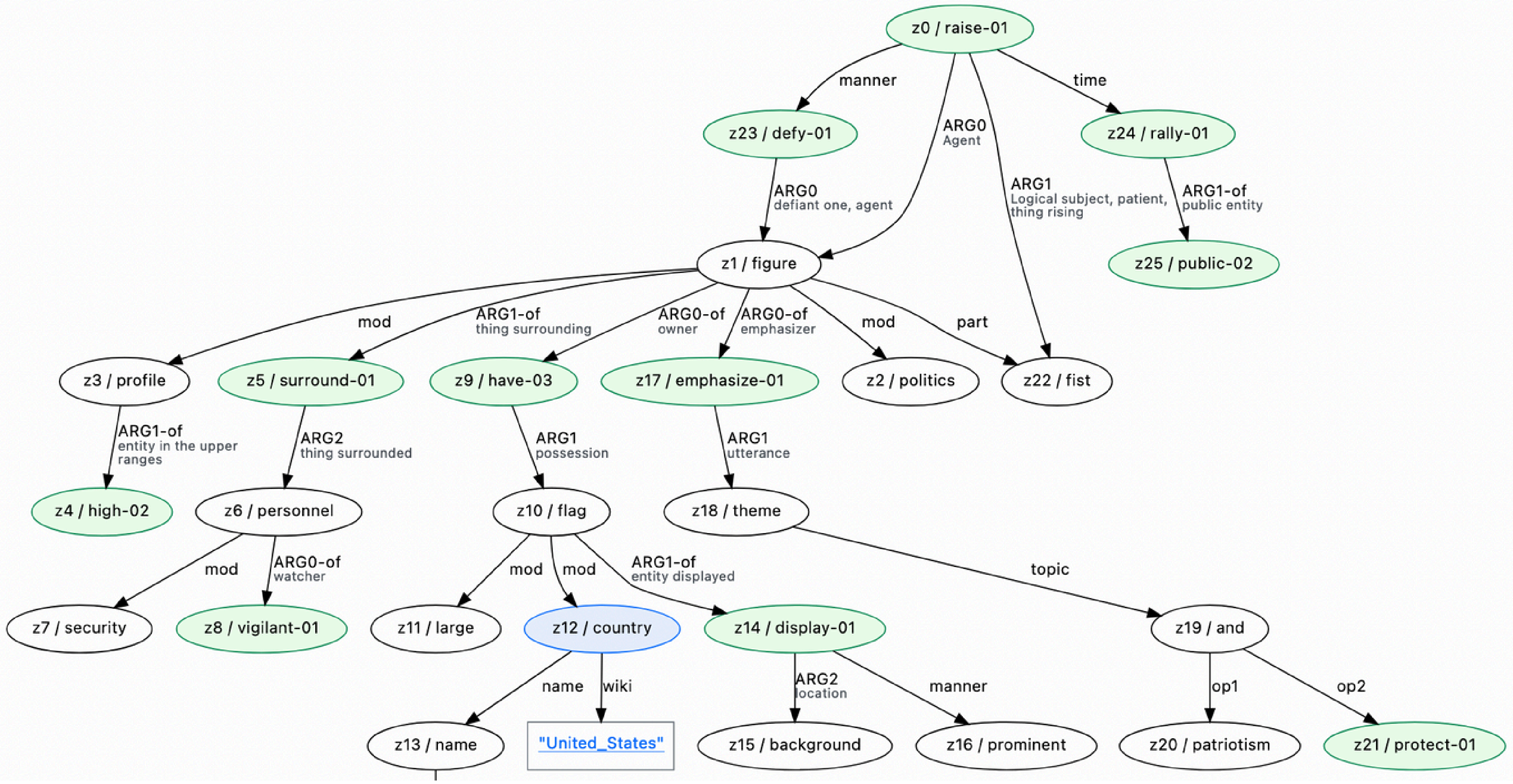
Text to AMR

Your text

PENMAN | Graph

Zoom: **double-click** to zoom-in and **Shift + Double-click** to zoom-out, or use **Shift + Scroll**

Pan: **drag left mouse button**



Penman and Graph notation for Text to AMR

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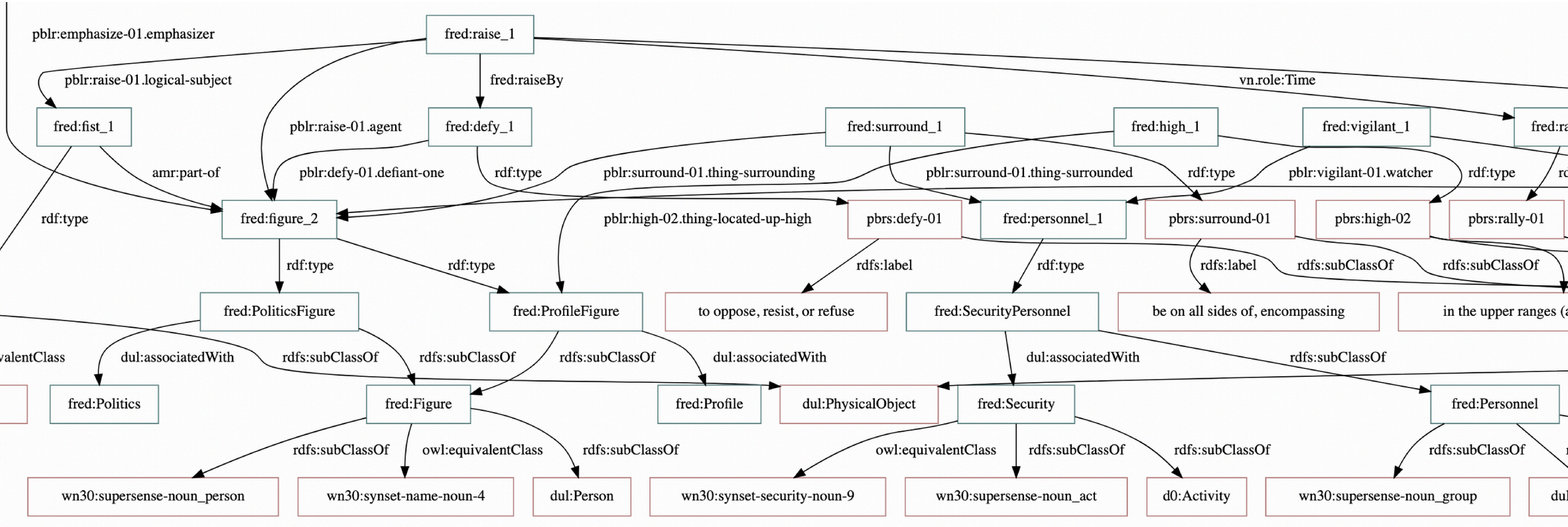
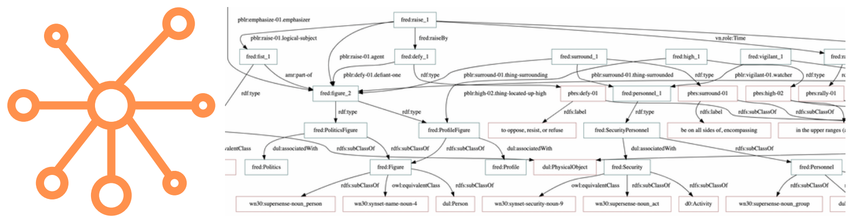
FRED graph



Knowledge Extension Pipeline - 3

Knowledge Graph



LLM

"Clever Prompting" Techniques

Semantic Layers for knowledge enrichment

```
You receive a text
"T"
and a frame-based
knowledge graph
"KG"
that is the extraction of
factual knowledge from T.

T:
{{ Text }}

KG:
{{ KG }}

Your goal is to extend KG
with more knowledge that
can be assumed from T, but
it is not explicit.
```

conversational_implicatures.prompt

```
Task:
Your goal is to extend KG with more knowledge that can be
assumed from T, but it is not explicit.
Using the elements of KG, and PropBank and WordNet elements as
linking points, add any further elements you need to extrat
implicit knowledge about:  Conversational implicatures.

Conversational implicatures, in the sense of Grice's
pragmatics.
Here are natural language inference examples:
1) She won't necessarily get the job -> She will possibly get
the job
...
```

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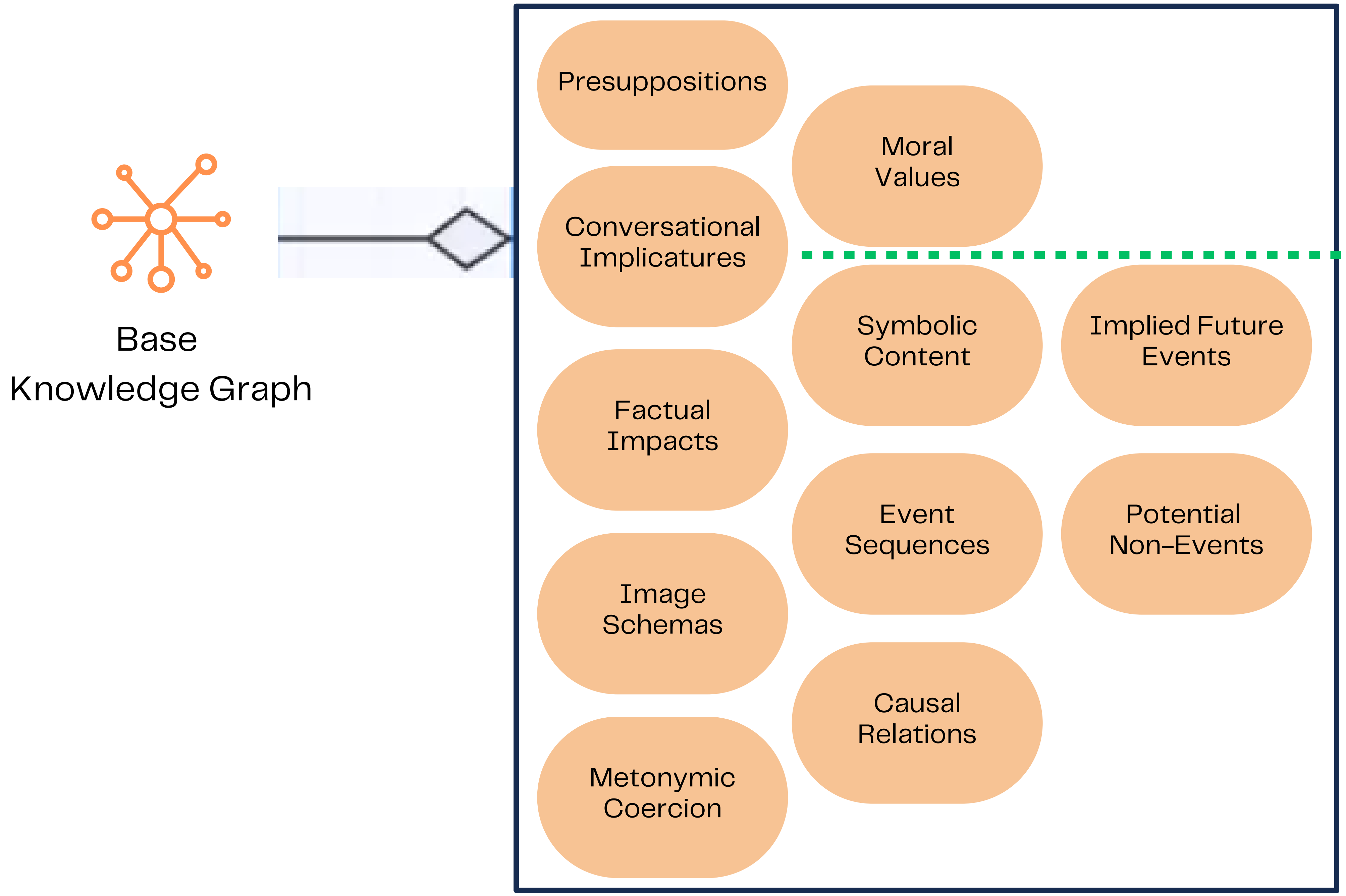
Knowledge Extension Pipeline - 4



"Clever Prompting" Techniques

Semantic Layers for knowledge enrichment

```
You receive a text
"T"
and a frame-based
knowledge graph
"KG"
that is the extraction of
factual knowledge from T.

T:
{{ Text }}

KG:
{{ KG }}

Your goal is to extend KG
with more knowledge that
can be assumed from T, but
it is not explicit.
```

conversational_implicatures.prompt

BASE GRAPH

```
fred:raise_1 a pbrs:raise-01 ;
    pblr:raise-01.agent fred:figure_2 ;
    pblr:raise-01.logical-subject fred:fist_1 .

fred:surround_1 a pbrs:surround-01 ;
    pblr:surround-01.thing-surrounded fred:personnel_1 ;
    pblr:surround-01.thing-surrounding fred:figure_2 .
```

The figure has a raised fist. The personnel surrounds the figure.

SYMBOLIC COERCION

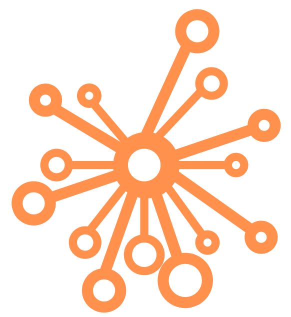
```
fred:fist_1 coerce:coercedType
    mor:Resistance, mor:Strength .

fred:flag_1 coerce:coercedType
    mor:NationalIdentity, mor:Patriotism .
```

The fist is a symbol of Resistance and strength.
The flag is a symbol of national identity and patriotism.

FACTUAL IMPACT

```
fred:figure_2 impact:hasExpectedEmotion impact:Defiance ;
    impact:hasExpectedSocialImpact impact:IncreasedPublicSupport .

fred:personnel_1 impact:hasExpectedEmotion impact:Tension ;
    impact:hasExpectedMentalState impact:Alertness .
```

The figure has expected social impact an increased public support.
The personnel has expected mental state alertness.



Extended K Graph

MORAL VALUES

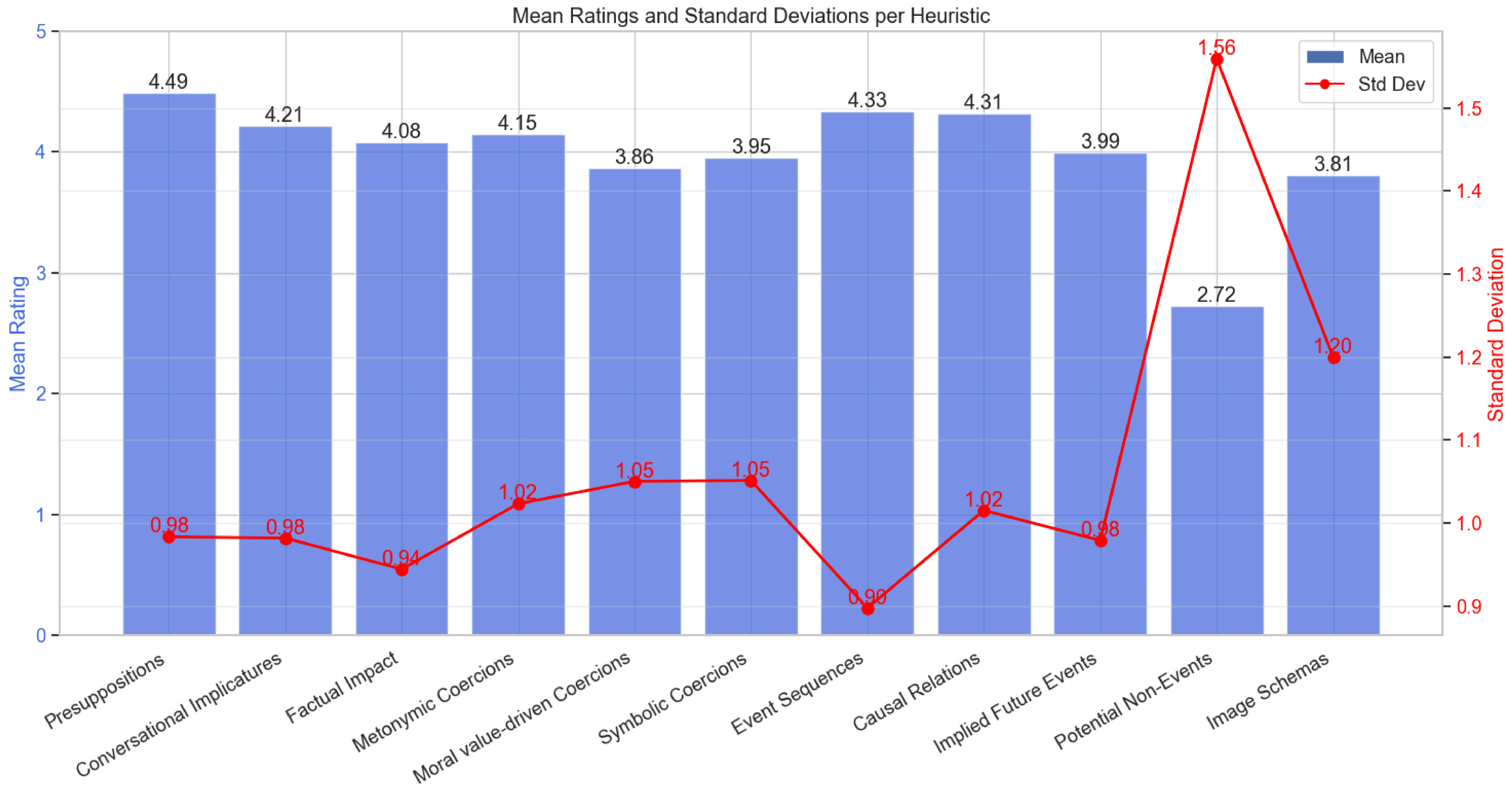
```
fred:assert_1 a pbrs:assert-02 ;
    mor:evokes mor:PowerDemonstration ;
    pblr:assert-02.agent fred:figure_2 ;
    pblr:assert-02.topic fred:power_1 .
```

The figure assertion is a demonstration of power.

IMPLIED FUTURE EVENTS

```
fred:support_1 a owl:ObjectProperty,
        pbrs:support-01 ;
    pblr:support-01.supported fred:figure_2 ;
    pblr:support-01.supporter fred:audience_1 .
```

We can expect a raise in the support from the audience.



Stefano De Giorgis, Aldo Gangemi, and Alessandro Russo. "Neurosymbolic graph enrichment for grounded world models." Information Processing & Management 62.4 (2025): 104127.

Event Segmentation

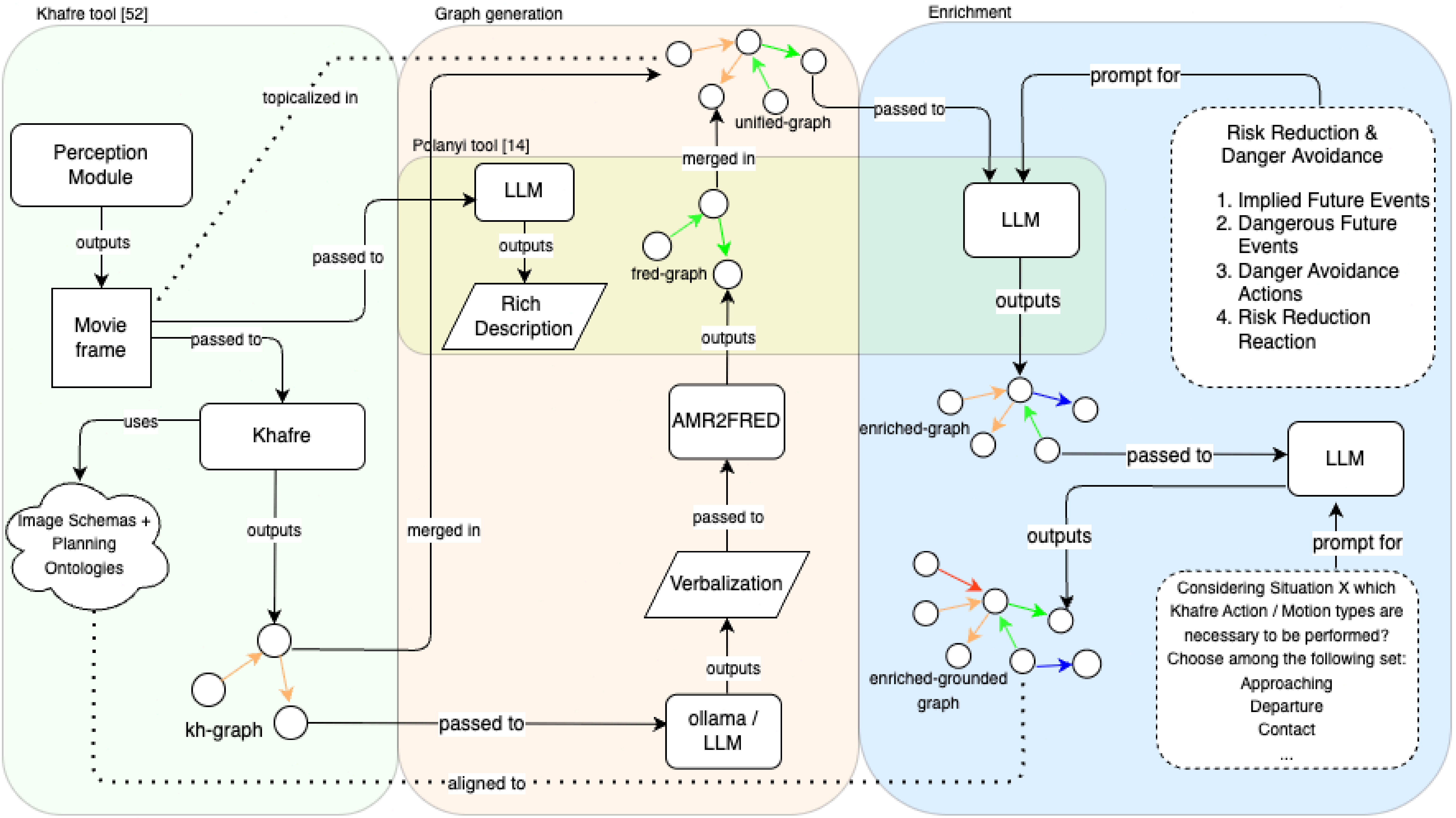
	0.12	0.15	0.51	0.54	0.57	0.75	0.81	0.96	0.99	1.02	1.05	1.11	1.17	1.86	1.95	2.34	2.73	2.91	3.33	3.42	3.54
Approach(hasParticipant: person_1,hasParticipant: person_2)	■	■	■	■	■	■	■	■	■	■	■										
Linkage(hasParticipant: person_1,hasParticipant: person_2)		■		■		■		■		■	■										
Contact(hasParticipant: person_1,hasParticipant: person_2)		■		■		■		■		■	■										
Departure(hasParticipant: cow_3,hasParticipant: person_1)		■	■	■	■	■	■	■	■	■	■	■	■			■	■	■	■		
Departure(hasParticipant: cow_3,hasParticipant: person_2)		■	■	■	■	■	■	■	■	■	■	■									
Stillness(hasParticipant: person_1,hasParticipant: sheep_3)											■	■	■	■							
Approach(hasParticipant: cow_3,hasParticipant: person_1)														■	■						■
Departure(hasParticipant: bird_3,hasParticipant: person_1)																	■		■	■	■
Approach(hasParticipant: bird_3,hasParticipant: person_1)																		■			
Stillness(hasParticipant: cow_3,hasParticipant: person_1)																				■	

```
log:hasParticipant rdf:type owl:ObjectProperty .
log:hasId rdf:type owl:DatatypeProperty .

log:image_0
    rdf:type owl:NamedIndividual ;
    rdf:type log:Image ;
    log:hasId "0.12"^^xsd:string .

log:schematicRelation_1
    rdf:type owl:NamedIndividual ;
    log:hasParticipant log:person_1 ;
    log:hasParticipant log:person_2 ;
    rdf:type log:Approach ;
    log:eventMode log:Started .

log:person_1 rdf:type owl:NamedIndividual ;
    rdf:type log:person .
log:person_2 rdf:type owl:NamedIndividual ;
    rdf:type log:person .
```

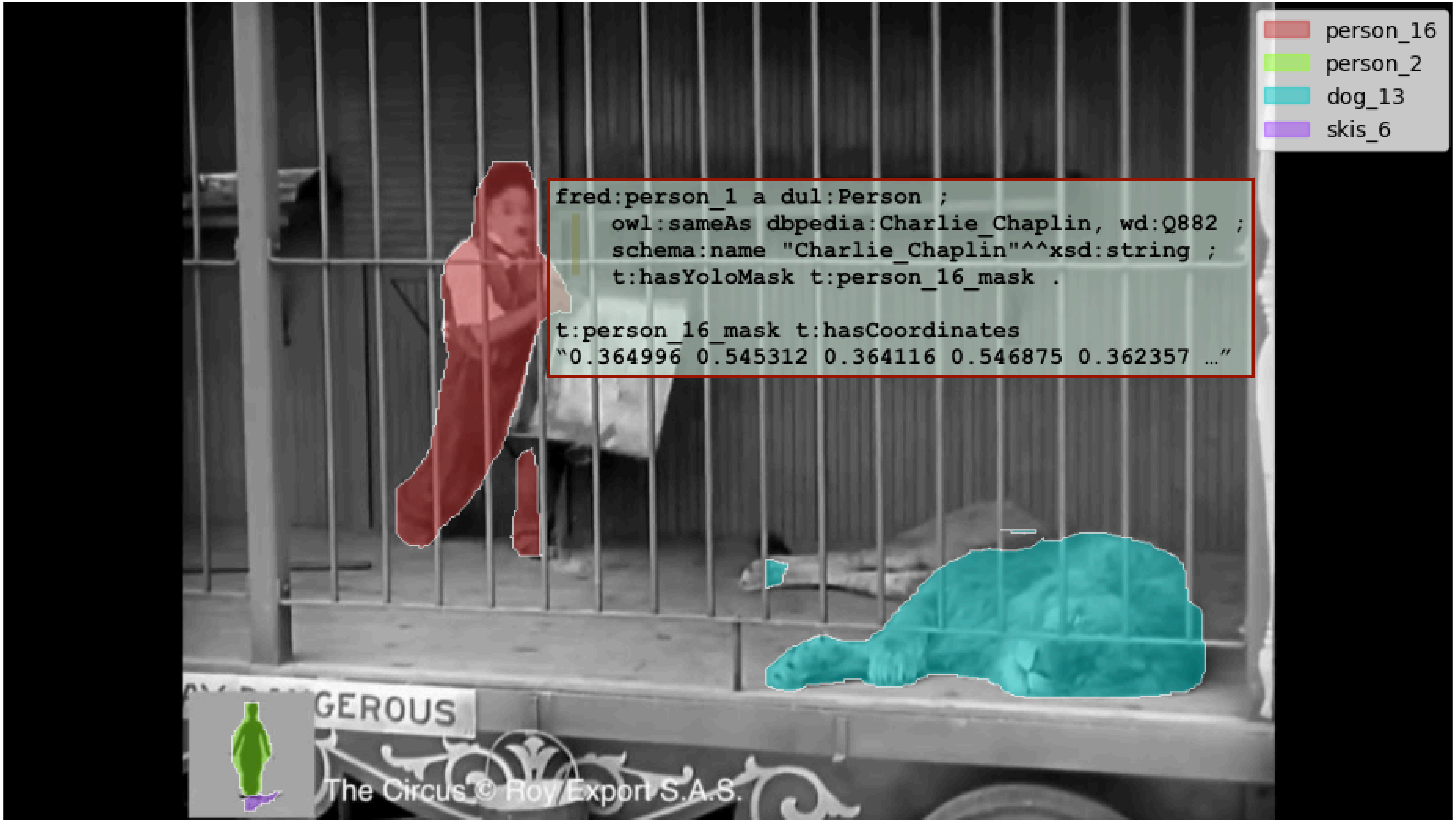
Neuro-symbolic knowledge enrichment - 2

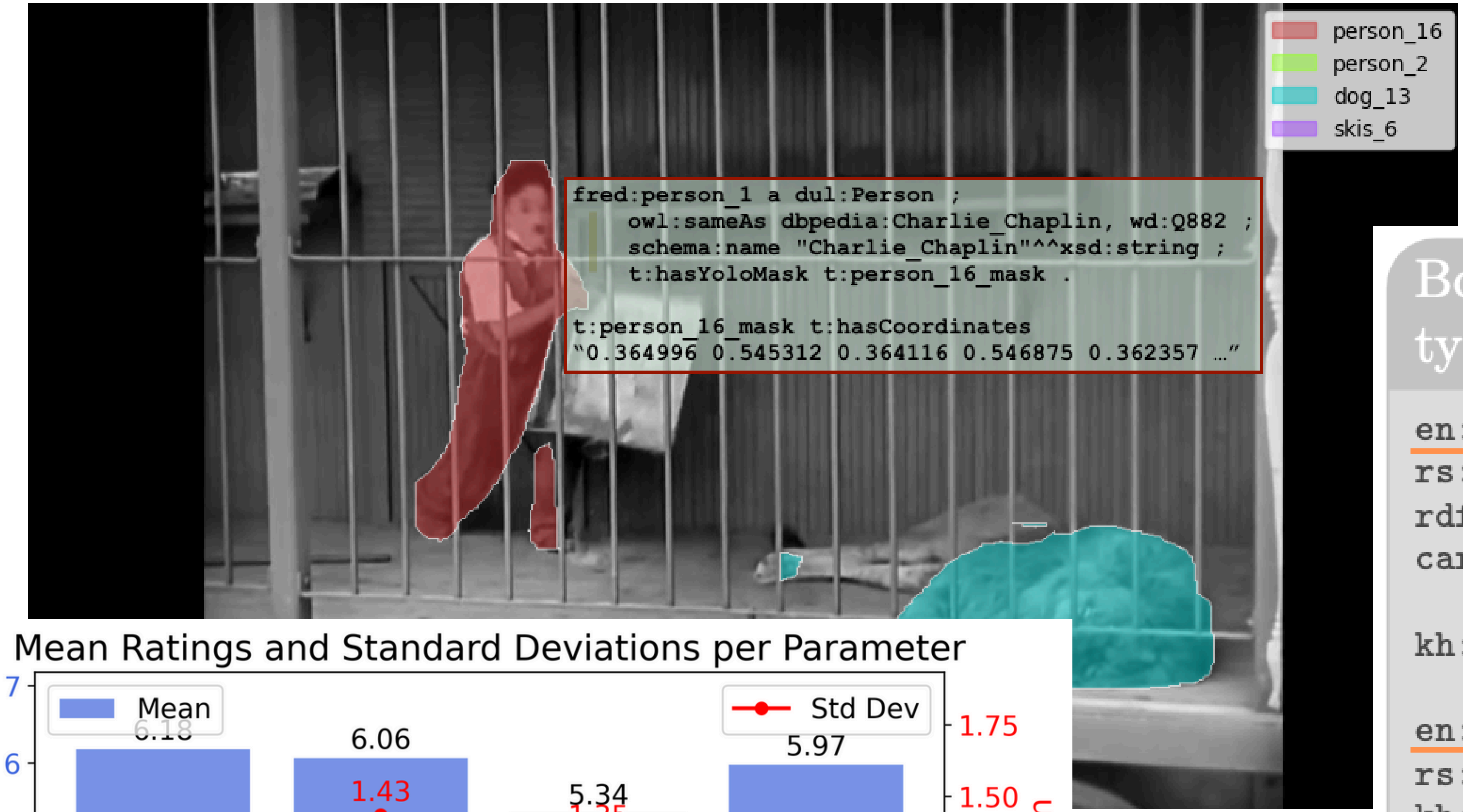
De Giorgis S., Righetti G., Pomarlan Hawkins M., Tsiogkas N., Hedblom M. Leemhuis M., Kutz O. - ongoing

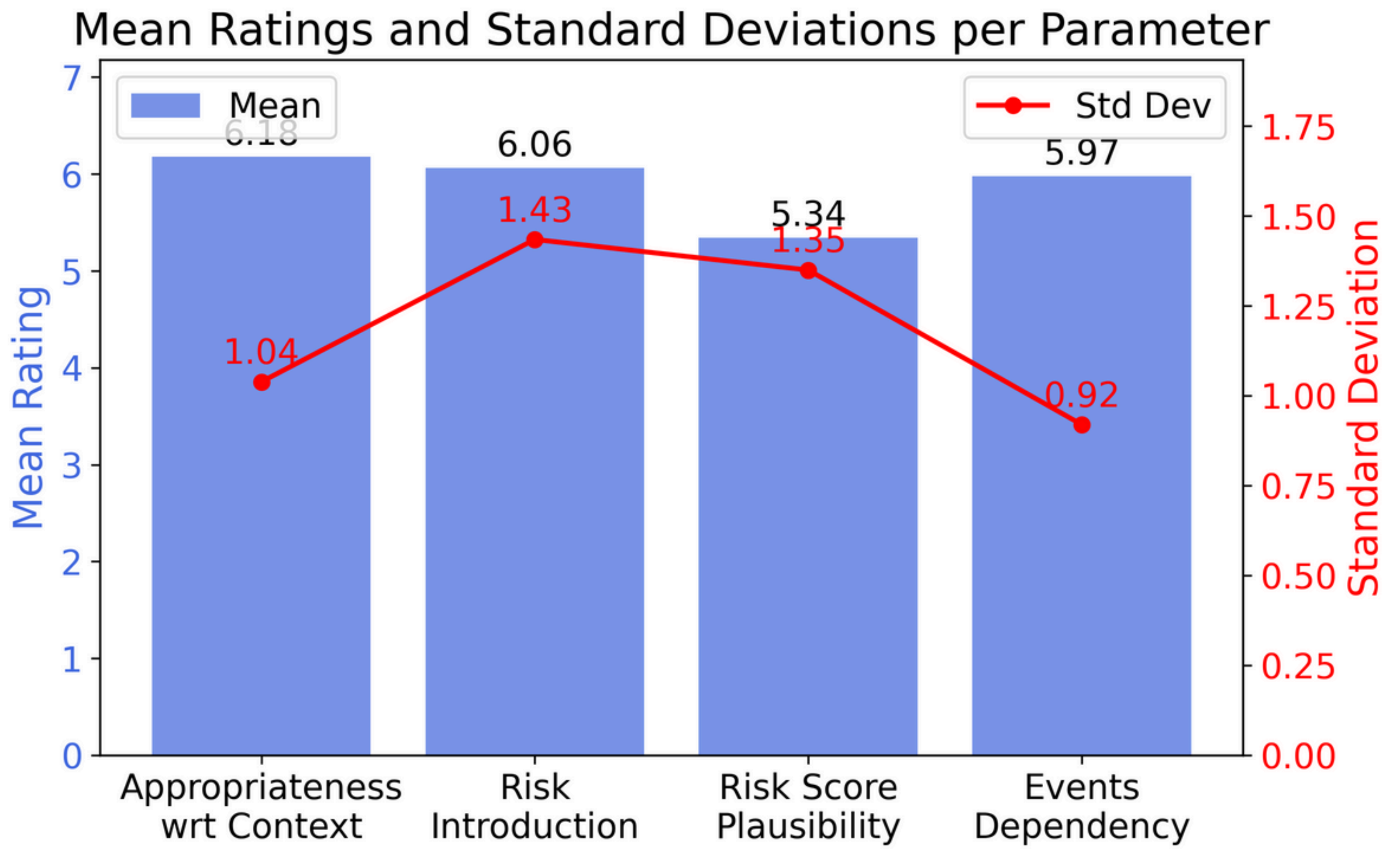
NEURAL COMPONENT – OBJECT RECOGNITION



Khafre tool [52]
Graph generation
Enrichment
topicalized in
Perception Module
outputs
Movie frame
passed to
passed to
Khafre
uses
Image Schemas + Planning Ontologies
outputs
kh-graph
passed to
Polanyi tool [14]
LLM
outputs
Rich Description
merged in
merged in
fred-graph
outputs
AMR2FRED
passed to
Verbalization
outputs
ollama / LLM
unified-graph
passed to
prompt for
LLM
outputs
Risk Reduction & Danger Avoidance
1. Implied Future Events
2. Dangerous Future Events
3. Danger Avoidance Actions
4. Risk Reduction Reaction
enriched-graph
passed to
LLM
outputs
prompt for
Considering Situation X which Khafre Action / Motion types are necessary to be performed? Choose among the following set:
Approaching
Departure
Contact
...
enriched-grounded graph
aligned to


person_16
person_2
dog_13
skis_6
fred:person_1 a dul:Person ;
owl:sameAs dbpedia:Charlie_Chaplin, wd:Q882 ;
schema:name "Charlie_Chaplin"^^xsd:string ;
t:hasYoloMask t:person_16_mask .
t:person_16_mask t:hasCoordinates
"0.364996 0.545312 0.364116 0.546875 0.362357 …"
GEROUS
The Circus © Roy Export S.A.S.





Box 6 - Recommended Action / Motion types

```
en:manExitingCageCarefully a dul:Situation ;
rs:requires kh:Movement, kh:Departure ;
rdfs:comment "The man should exit the cage
carefully without making sudden movements" .

kh:person_16 dul:hasRole kh:Mover, kh:Departer .

en:manStumblingBackward a dul:Situation ;
rs:requires kh:Movement, kh:Departure,
kh:Falling ;
rdfs:comment "The man might stumble while moving
backward, drawing the lion's attention" .

kh:person_16 dul:hasRole kh:Mover, kh:Departer,
kh:Faller .

kh:person_16 dul:hasRole kh:Mover, kh:Departer .
kh:dog_13 dul:hasRole kh:ReferencePoint .
```

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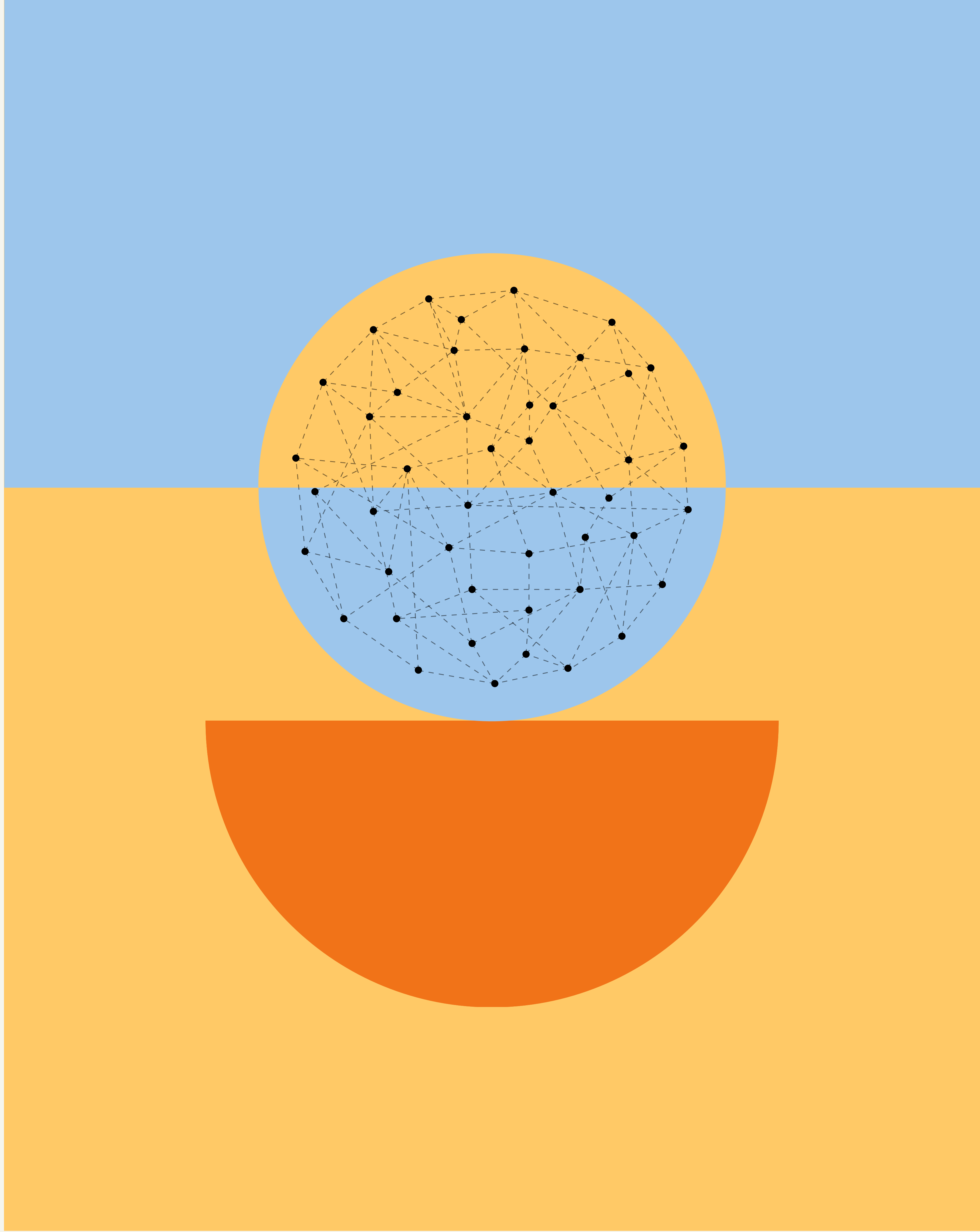
RECOMMENDED READING

Back-up slides

— RELATED WORK



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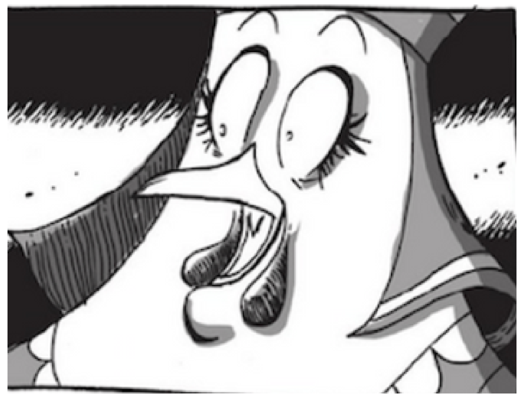

STLAB
SEMANTIC TECHNOLOGY LABORATORY

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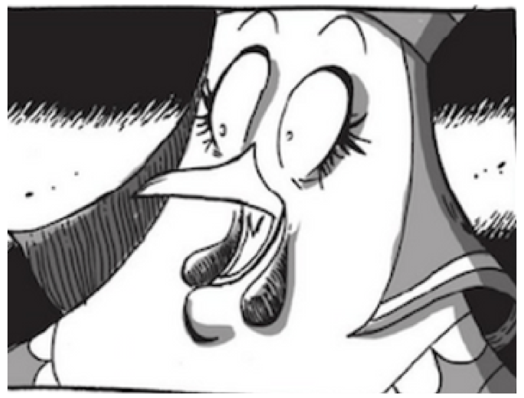

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Post-doc at Vrije Universiteit Amsterdam with prof. Filip Ilievski and prof. Frank van Harmelen on the "Interpretation of Internet Memes"